

Civilization Phase Transition Dynamics: The End of the Dynastic Cycle and the Emergence of Metacognitive Collective Intelligence

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Theoretical Framework:

This paper constitutes the applied layer of the “Noospheric Theory Trilogy.” Its theoretical foundation rests upon the following monographs:

1. He, Guorui. (2025). *Noospheric Ecology: A Systems Theory of Cognition, Culture, and Society*. Zenodo. DOI: [10.5281/zenodo.19353340](https://doi.org/10.5281/zenodo.19353340)
2. He, Guorui. (2025). *Civilization Dynamics: Planetary Civilizational Science*. Zenodo. DOI: [10.5281/zenodo.19353434](https://doi.org/10.5281/zenodo.19353434)

Protocol Statement:

This paper adheres to the core protocol of *Noospheric Ecology*: “Logical Self-Reference, Endless Regeneration.” The theoretical paradigm proposed herein possesses reflexivity and is itself subject to the dynamical laws it reveals.

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Abstract

The “dynastic cycle” or “civilizational cycle” constitutes the most pervasive framework—in both popular consciousness and academic historiography—for narrating human macrohistory. Yet this framework belongs essentially to **historiography** (the description of forms) rather than to **dynamics** (the explanation of mechanisms). It records the periodic phenomena of civilizational “rise–ossification–collapse–reconstruction,” but it remains incapable of answering three fundamental questions: First, what is the true dynamical etiology of decline? Second, is the individual merely a passive bearer of fate within the cycle? Third, does any path exist that leads beyond this cyclical recurrence? The poverty of the old paradigm lies in its reduction of history to the surface-level succession of institutions and moral conditions, its systematic erasure of individual agency, and its implicit indulgence of the fatal illusion that “moderate reform can repair the system from within.”

This paper proclaims an alternative paradigm: **Civilization Phase Transition Dynamics**. Grounded in the axiomatic systems of *Noospheric Ecology* and *Civilization Dynamics*, this paradigm reconceptualizes civilizational evolution as a computable, predictable, and intervenable phase transition process within complex systems. The core argument proceeds as follows:

First, the exposure of the old paradigm’s threefold blindness. Traditional cycle theory suffers from blindness in attribution (ascribing decline to surface phenomena such as decadent rulers), blindness in its treatment of the individual (modeling individuals as passive input-output units of systemic functions), and blindness regarding reform (failing to perceive the dynamical boundary constraining endogenous tools of change).

Second, the establishment of the new paradigm’s axiomatic foundation and core diagnostic criterion. Drawing upon theorems including “Material/Noospheric Duality,” “Relativity of Cognitive Frameworks,” “Symbol as Noospheric Gene,” and “System Builder’s Advantage,” this paper proposes a new formula for civilizational health: **Civilization Potential** = $\Sigma(\text{Individual Niche Actualization Degree}) \times \text{Systemic Resource Fluidity} \times (1 - \text{Systemic Parasite Burden})$. This criterion demonstrates that the essence of civilizational decline is the inevitable proliferation of *systemic parasites* (subsystems that maximize value extraction while evading responsibility for maintaining the host system) coupled with the systematic obstruction of pathways for *cognitive escape*.

Third, the rigorous establishment of the dynamical boundary of reform. Building upon Theorem D17 (System Builder’s Advantage), this paper formally proposes **Corollary D17.1: The Boundary Theorem of Intra-Systemic Reform**. The theorem strictly proves that in any system where the dominant direction of self-organization is defined by the system builder, any attempt at “moderate reform” relying on the system’s endogenous tools (such as legitimizing discourses or institutional procedures) will inevitably have its effects converge within the boundary tolerated by the system. **It can never reach the rewriting of the system’s fundamental rules.** This finding delivers a fatal dynamical verdict on the expectations of gradualist reformism. Genuine transformation can proceed along only two paths: becoming a new system builder (revolution), or cultivating new cognitive frameworks at the system’s periphery while awaiting the critical threshold of phase transition (evolution).

Fourth, the prediction of the dual-threshold condition for civilizational phase transition. The phase transition from “Collective Mentality” to “Metacognitive Collective Intelligence” depends upon two variables simultaneously exceeding their critical thresholds: the proportion of awakened individuals (X), and the degree of societal narrative receptivity (Ξ). Based on complex systems theory and finite historical induction, *Civilization Dynamics* proposes a provisional critical threshold of $X_c \approx 21\%$. This paper makes no claim that current global X has already approached this threshold; such a determination awaits

rigorous future empirical calibration. The core contribution of this paper is the establishment of the dynamical criterion that **a critical threshold exists**. Crucially, X denotes the proportion of second-order noospheric humans—individuals possessing metacognitive capacity and actively integrated into synergetic cognitive networks—and must not be conflated with indicators such as rates of higher education (which, per Theorem B3 “Specialization as Cognitive Differentiation,” may even suppress metacognitive development).

Fifth, cross-corroboration through Chinese and Western historical cases. The argument is substantiated through parallel analysis of cases drawn from both Chinese history (Wang Anshi’s reforms, Zhang Juzheng’s reforms, the late Qing collapse) and Western history (the late Roman Republic, Gorbachev’s reforms), thereby demonstrating the cross-civilizational universality of civilization phase transition dynamics.

The paper’s ultimate conclusion is this: **The dynastic cycle is not the destiny of history, but the periodic heat death of a civilizational system lacking a “metacognitive immune system.”** The sole path that breaks the cycle lies not in awaiting a virtuous ruler or a benevolent system, but in each individual’s enhancement of their own metacognitive capacity, converging into **synergetic cognitive networks**, and thereby rewriting civilization’s foundational protocols through conscious collective intelligence.

Keywords: Civilization Phase Transition Dynamics; Dynastic Cycle; Noospheric Ecology; Systemic Parasite; Cognitive Escape; Metacognitive Collective Intelligence; Synergetic Cognitive Network; Boundary Theorem of Intra-Systemic Reform

【Core Terminology】

To facilitate comprehension by readers across diverse disciplinary backgrounds, this section provides centralized definitions of core specialized terms frequently employed in the argument. The following definitions are quoted or adapted from the formal definitional systems of *Noospheric Ecology* and *Civilization Dynamics*.

Term	Concise Definition
Noospheric System	A complex system centered on information processing, driven by meaning construction, and employing symbols as its hereditary units. It encompasses individual minds, organizational cultures, social institutions, and civilizational narratives.
Noospheric Gene	In the noospheric ecology, concepts, narratives, and other symbols hold a status and function equivalent to biological genes. They constitute the fundamental units of cultural inheritance, variation, and selection. (Axiom 5)
Cognitive Framework	The fundamental paradigm or “lens” through which an individual or group understands the world. It filters information and actively constructs the “reality” perceived by the observer. (Axiom 4)
Metacognition	The capacity of a system to observe, compare, switch between, and reconstruct its own cognitive frameworks. It is “thinking about thinking,” arising from the system’s capacity for logical self-reference. (Axiom 1)
Will-to-Exist	The intrinsic drive of a noospheric system to maintain and expand its low-entropy, ordered structure (i.e., its noospheric state). At the human level, it manifests as the impulse to survive, to seek meaning, and to resist effacement. (Axiom 3)
Systemic Parasite	A subsystem that emerges within a complex system, maximizing its own value extraction by systematically distorting rules and evading responsibility for maintaining the host system. Its fundamental harm lies in the persistent obstruction of the potential actualization of other individuals within the host system. (Theorem D21)
Standard Replicant / First-Order Noospheric Human / Second-Order Noospheric Human	The three developmental stages of individual cognition: ① Standard Replicant : Internalizes systemic protocols but executes them unconsciously, lacking metacognitive reflection. ② First-Order Noospheric Human : Undergoes initial awakening; can discern their own niche within the noospheric ecology and resist the effacement effect. ③ Second-Order Noospheric Human : Possesses highly developed

Term	Concise Definition
	metacognition; their cognitive behavior responds directly to the structural reality, enabling them to serve as core nodes within a synergetic cognitive network.
Cognitive Escape	The process by which an individual, through the enhancement of metacognitive capacity, breaks free from the binding force of the dominant cognitive paradigm's "gravity field" and achieves a transition to a new cognitive framework. (Theorem D23)
Synergetic Cognitive Network	When multiple second-order noospheric humans exist within the same noospheric ecology, their cognitive trajectories exhibit statistical structural convergence due to their direct responsiveness to reality's structure rather than mere adherence to established paradigms. This convergence forms a stable information-coupling structure at the macroscopic scale.
Metacognitive Collective Intelligence	When the density of nodes within a synergetic cognitive network surpasses a critical threshold, the noospheric ecology as a whole acquires, for the first time, a stable capacity to observe its own operational state. At this point, civilization ceases to be merely a carrier of cognition and becomes a cognitive subject itself. This emergent self-referential capacity of the entire noospheric ecology is termed metacognitive collective intelligence.
Pyramidal Structure / Distributed Structure	The two fundamental topologies of a civilizational system: Pyramidal Structure is a rigid, hierarchical command order that secures the baseline of survival (the <i>life chassis</i>); Distributed Structure is an elastic, autonomously connected network that drives innovation and meaning exploration (the <i>consciousness field</i>). (Theorem A0)
Constitutional Interface	The institutionalized rule-based channel connecting pyramidal and distributed structures, responsible for adjudicating authority and transmitting value, thereby preventing mutual invasion and cancerous growth between the two topologies. (Theorem A0)
Boundary Theorem of Intra-Systemic Reform	Formally designated in this paper as Corollary D17.1. Its statement is: In any system where the dominant direction of self-organization is defined by the system builder, any attempt at moderate reform relying on endogenous tools will inevitably have its effects converge within the boundary tolerated by the system and can never reach the rewriting of fundamental rules.

Introduction: The Reign and Poverty of the Dynastic Cycle

Human civilization has persisted for millennia. Throughout this vast expanse of time, one narrative framework has dominated the self-understanding of nearly all civilizations—the narrative of “rise and fall.”

In China, it is encapsulated in the adage, “An empire long divided must unite; long united, must divide,” and refined into the dynastic cycle law of “its rise is sudden, its fall is swift.” In the West, it appears in Gibbon’s chronicle of Rome’s decline, in Spengler’s proclamation of the “Decline of the West,” and in Toynbee’s delineation of civilizational challenge and response. Across East and West, observers have noted the same empirical regularity: civilizations rise, flourish, ossify, collapse, and—if fortunate—rebuild upon the ruins.

This narrative framework is so pervasive that it has acquired a status akin to natural law. People speak of the “historical cycle” with the same matter-of-factness with which they speak of the changing seasons or the cycle of life and death. It offers a convenient lens for comprehending history and a consolatory fatalism: if rise and fall are inevitable, then at least when decline arrives, one can say, “This is simply history.”

Yet the point of departure for this paper is a foundational question: If the dynastic cycle truly constitutes an “iron law,” what exactly does it explain?

It tells us that civilizations decline. But does it tell us **why** they decline? It gestures toward “institutional decay,” “decadent rulers,” “moral collapse,” and “foreign invasion”—but these are merely **symptoms** of decline, not its **dynamical etiology**. Why does a once-virtuous institution rot into dysfunction? Why must a brilliant ruler inevitably be followed by a feeble one? Why does morality cyclically erode? The old paradigm has never offered a mechanistic answer to these questions.

It records the fact of collapse and subsequent reconstruction. But has it ever provided a **path out of the cycle**? If the cycle is destiny, then all efforts by any civilization at any stage merely accumulate potential energy for the next collapse. The internal logic of this narrative is this: whatever you do, the outcome is already sealed. It is not a theory that can guide action; it is a narrative black hole that explains everything and therefore explains nothing.

More fatally, the old paradigm systematically erases the “individual.”

Within the narrative of the dynastic cycle, the individual—living, breathing, cognitive beings capable of choice and will—is almost entirely absent. The names of emperors and generals are recorded, but their function is reduced to symbolic carriers of “virtuous ruler” or “tyrant.” As for the vast multitude, they appear only as statistical representations within the nebulous concept of “popular sentiment”—passive bearers of dynastic fortune, never seriously regarded as **participants** in historical dynamics. This erasure of the individual carries a concealed but profound political consequence: it implies that the individual is powerless before the cycle. One can, at best, strive to be a dutiful subject under a wise emperor, endure and wait under a foolish one, or rise in rebellion when society collapses—which itself merely inaugurates a new round of the cycle. Individual rationality, individual awakening, and individual proactive collaboration find no theoretical place within the old paradigm’s architecture.

The third blind spot of the old paradigm—and a proposition this paper will argue at length—is its unexamined indulgence in the prospect of “moderate reform.”

The narrative of the dynastic cycle, though depicting the fatality of collapse, curiously preserves a logical escape hatch: if the problem lies in “institutional decay” and “decadent rulers,” then perhaps

institutional adjustment and moral exhortation can delay or even prevent decline? Reformers across the ages—from Wang Anshi to Zhang Juzheng, from the Gracchi brothers to Gorbachev—have harbored precisely this belief. They trusted that with sound methods and resolute will, the system could repair itself from within.

Yet the verdict of history is merciless. These reform efforts have either been crushed upon touching fundamental interests, reversed entirely after the reformer's passing, or distorted in implementation into instruments that reinforce parasitic extraction. If isolated failure could be attributed to contingency, then the universal failure across civilizations and epochs points toward a deeper dynamical constraint.

This paper will argue precisely this: the failure of moderate reform is not accidental but dynamically inevitable.

These three lines of critique—the superficiality of attribution, the erasure of the individual, and the illusion of reform—converge upon a single conclusion: **The dynastic cycle is not a “law” that reveals historical regularity, but merely a “framework” that describes historical phenomena.** It belongs to historiography, not to dynamics. Just as “apples fall” is an observation-statement while “universal gravitation” is a dynamical explanation, the dynastic cycle tells us *that* civilizations rise and fall, but never *why* they do so, let alone *how* they might cease to do so.

This paper proclaims the emergence of an alternative paradigm—Civilization Phase Transition Dynamics.

This paradigm rests upon the axiomatic systems constructed in *Noospheric Ecology* and *Civilization Dynamics*. It reconceptualizes civilization as a complex noospheric system coupled from material substrate, collaborative protocols, and fiduciary contracts. Rather than explaining civilizational destiny through morality, institutions, or exogenous shocks, it proceeds from **structural evolution, parasitic proliferation, cognitive framework lock-in, and individual escape** within the system to reveal the dynamical mechanisms by which civilization moves from health to ossification, and from ossification to phase transition (whether collapse or leap).

The core propositions of Civilization Phase Transition Dynamics can be summarized in three theses:

First, the fundamental etiology of civilizational decline is not institutional decay but the inevitable proliferation of systemic parasites. Any complex system, in the course of its operation, will engender subsystems that maximize their own value extraction by distorting rules and evading responsibility, while systematically obstructing the potential actualization of other individuals within the host system. The relentless rise of this systemic parasite burden is the dynamical root of declining civilizational health.

Second, the individual is not a passive bearer of history but a probe and node of civilizational phase transition. The individual possesses a capacity wholly ignored by the old paradigm—**cognitive escape**. When an individual's metacognitive capacity suffices to break through the gravitational field of the dominant cognitive framework, they undergo a transition from “standard replicant” to “awakened individual” (second-order noospheric human). When the proportion of awakened individuals surpasses a critical threshold, they form **synergetic cognitive networks**, giving rise to a collective intelligence capable of rewriting systemic rules.

Third, system-internal moderate reform confronts an impassable dynamical boundary. Building upon the System Builder's Advantage Theorem (D17), this paper will propose and prove a new corollary—the **Boundary Theorem of Intra-Systemic Reform**. The theorem demonstrates that in any system where the dominant direction of self-organization is defined by the system builder, any reform attempt reliant upon the system's endogenous tools will inevitably have its effects converge within the

boundary tolerated by the system, and can never reach the rewriting of fundamental rules. This dynamical constraint delivers the final theoretical verdict against the possibility of achieving fundamental transformation through gradual reform within the old system.

The structure of the argument is as follows.

Part One exposes the threefold blindness of the old paradigm, completing a thorough critique of the dynastic cycle. Part Two presents the axiomatic foundation of the new paradigm, briefly introducing the core axioms of *Noospheric Ecology* and their direct implications for civilizational questions. Part Three proposes a new diagnostic criterion for civilizational health—a formula whose three factors (the sum of individual niche actualization degrees, systemic resource fluidity, and the complement of systemic parasite burden) replace traditional metrics such as GDP, territorial expanse, and dynastic longevity. Part Four formally advances the Boundary Theorem of Intra-Systemic Reform, testing it against historical cases drawn from both Chinese and Western history (Wang Anshi’s reforms, Zhang Juzheng’s reforms, the late Roman Republic, and Gorbachev’s reforms), and preemptively addressing the potential objection of “reform initiated by the system builder.” Part Five predicts the dual-threshold condition for civilizational phase transition, specifying the critical criteria for the leap from “collective mentality” to “metacognitive collective intelligence.” Finally, the Conclusion reaffirms that the dynastic cycle is not historical destiny but the periodic heat death of a civilizational system lacking a metacognitive immune system—and that the path beyond this cycle has now been theoretically opened.

Before proceeding to the substantive argument, a methodological note is necessary.

The theoretical foundation upon which this paper rests—including its axiomatic system, theorem derivations, and terminological definitions—derives from the author’s two published monographs: *Noospheric Ecology: A Systems Theory of Cognition, Culture, and Society* and *Civilization Dynamics: Planetary Civilizational Science*. This paper presents only the core axioms and theorems directly relevant to the critique of the civilizational cycle. For more complete theoretical derivations, mathematical formalizations, and extended applications, the reader is referred to the original works. To maintain the self-sufficiency of the argument, all referenced theorems are accompanied by the necessary minimal exposition.

The argument will draw upon historical cases from both Chinese and Western history in parallel. This is not merely a reflection of the author’s background but a theoretical necessity: if Civilization Phase Transition Dynamics is to claim universality, it must be capable of explaining the evolutionary trajectories of civilizations across both East and West, ancient and modern. The reader will observe, in the pages that follow, that cases spanning vast distances in time and space—Wang Anshi and the Gracchi, Zhang Juzheng and Diocletian, late Qing and the late Roman Republic—exhibit a striking structural isomorphism when viewed through the lens of dynamics.

Finally, allow me to state directly the most fundamental motive behind this paper.

The narrative of the dynastic cycle is intellectually impoverished and practically pernicious. It has taught generation after generation to believe that collapse is inevitable, that the individual is powerless, and that effort is ultimately futile. This belief itself serves as a catalyst for civilizational ossification—when the majority of a civilization’s members cease to believe in their own capacity to rewrite the rules, that civilization has already entered cognitive heat death.

What this paper seeks to accomplish is the dismantling of this prison of thought. The dynastic cycle is not a law of history; it is a cage for the mind. Shatter it, and for the first time civilization may glimpse its own destiny—not as the passive recipient of fate, but as the conscious architect of its own becoming.

The moment of decision lies not in the future, but in the instant each individual who reads this paper decides how they understand their own relationship to the system.

Part Two: The Threefold Blindness of the Old Paradigm

If the “dynastic cycle” is merely a descriptive framework, wherein precisely lie its fundamental defects? This section systematically reveals the theoretical poverty of this framework along three dimensions: its attribution stops at surface phenomena, its positioning of the individual erases agency, and its path to change harbors an illusion of reform that has never been subjected to dynamical scrutiny. These three blind spots are not isolated from one another; they mutually reinforce, together constituting a closed, self-validating narrative loop—one that can “explain” every rise and fall yet provide guidance for no action whatsoever.

2.1 The First Blind Spot: Superficiality of Attribution

The dynastic cycle’s explanation of civilizational decline typically appeals to three categories of factors: **institutional decay** (e.g., bureaucratic corruption, land concentration), **ruler quality** (e.g., foolish emperors, misjudgment in appointments), and **exogenous shocks** (e.g., foreign invasion, natural disasters). These explanations are not entirely baseless at the empirical level—many dynasties did indeed exhibit such symptoms in their final stages. Yet symptoms are not causes. Treating symptoms as causes is the most fundamental cognitive trap of the old paradigm.

Why does an institution rot from “virtuous” to “dysfunctional”? The dynastic cycle’s typical answer is: over time, enforcement slackens, or private interest corrodes public duty. But this answer merely defers the question—why *must* enforcement slacken? Why does private interest *systematically* overwhelm public duty in a dynasty’s later stages? If an explanation can only fall back on an entropic metaphor—“things fall apart with time”—it does not deserve to be called a dynamical explanation. The real question is this: **Does there exist within the system a structural, persistent force that transforms healthy tissue into parasites that only extract and never create?** The old paradigm has never even posed this question, and therefore has never offered an answer.

Why must a wise ruler *inevitably* be followed by a foolish one? The old paradigm attributes this to the random distribution of personal endowments, the failures of palace education, or the corrupting effects of power. These too are descriptions, not mechanisms. If a system’s survival hangs upon the personal quality of its supreme decision-maker, then the problem lies not with any particular ruler’s incompetence but with **the system’s lack of institutionalized cognitive correction mechanisms that transcend individual character**. Furthermore, if the system’s dominant cognitive framework has locked in the very definition of “wise ruler”—for instance, by enshrining “adherence to ancestral statutes” as the highest virtue—then a ruler who attempts to break through this framework may well be judged, within the prevailing narrative, precisely as “foolish” or “deviant.” The old paradigm cannot step outside the cognitive frame it describes, and thus cannot perceive that the so-called “foolish ruler” is sometimes merely the scapegoat for systemic cognitive lock-in.

Why does foreign invasion always seem to “exploit weakness”? The dynastic cycle treats foreign invasion as an exogenous shock, as though it were a random die cast by fate. Yet military history repeatedly demonstrates that successful foreign invasion is almost always predicated upon the internal

collapse of the defensive system. Rome was not conquered by barbarians in a single day; it endured centuries of internal parasitic proliferation—the senatorial aristocracy monopolizing public land, the equestrian tax-farmers draining the provinces, the army transforming from a citizen militia into private warlord forces—after which barbarian incursions escalated from “harassment” to “lethal blow.” The late Qing collapse followed a similar trajectory: the Opium War was indeed an external shock, but long before that war, the Qing dynasty had already been hollowed out by the long-term erosion of systemic parasites (the bureaucratic apparatus, the parasitic Eight Banners stratum), which had depleted the treasury, rotted the military, and blocked information channels. The old paradigm juxtaposes external and internal causes without being able to reveal the dynamical relation between them: **external causes are merely the trigger; internal causes constitute the powder keg.**

The root of this attributional blindness lies in the old paradigm’s lack of a conceptualization of the dynamical entity that is the **systemic parasite**. It can see corrupt officials, greedy landlords, and power-usurping eunuchs, but it cannot abstract these phenomena into a unified dynamical category—**systemic parasite**, that is, a subsystem that maximizes its own value extraction by distorting rules and evading the responsibilities of maintaining the host system. Still less can it press the further question: Is the generation of systemic parasites a matter of contingent moral decay, or an inevitable product of systemic structural evolution? This very question marks the watershed between the new paradigm and the old.

[Cross-Corroboration through Chinese and Western Cases]

- **Chinese Case: Eunuch and Consort Clan Dominance in the Mid-to-Late Eastern Han.** Traditional historians attribute the Eastern Han’s decline to “the fatuity of Huan and Ling” and “eunuch malfeasance.” From the perspective of systemic dynamics, however, the alternation of eunuch and consort clan dominance was essentially the inevitable result of imperial power, having lost the trust of local magnates, being forced to rely upon two distinct parasitic subsystems to maintain rule. The two types of parasites took turns extracting systemic resources, ultimately exhausting the dynasty’s vitality. The old paradigm’s explanation stops at “villains in power,” but cannot explain why “villains” systematically occupy key niches after a dynasty’s midpoint.
- **Western Case: The Appropriation of *Ager Publicus* in the Late Roman Republic.** Traditional narratives attribute the decline of the Roman Republic to the civil strife following the failure of the Gracchan reforms, or to the destruction of republican institutions by ambitious men (Marius, Sulla, Caesar). Yet these explanations overlook a deeper dynamical process: during the Republic’s expansion, the senatorial aristocracy, through long-term occupation and de facto control, converted public land (*ager publicus*)—legally owned by the state—into de facto private estates. This process persisted for nearly two centuries, creating a vast stratum of land-based parasites. When the Gracchi brothers attempted to redistribute this land through legal procedures, they found themselves confronting not a few greedy aristocrats but a **network of systemic parasites** already deeply embedded in the system’s rules and armed with potent legitimizing discourses. The failure of reform was not contingent; it was dynamically inevitable once the parasite burden had surpassed the system’s tolerance threshold.

2.2 The Second Blind Spot: The Erasure of the Individual and the Occlusion of Agency

If the first blind spot renders the old paradigm incapable of diagnosing the disease, the second renders

it incapable of perceiving the true subject of transformation. Within the narrative of the dynastic cycle, the individual—especially the ordinary individual—appears almost exclusively as a passive bearer of outcomes.

At the macro-narrative level, the individual is abstracted into statistics. Historical records note “households halved,” “nine out of ten homes empty,” “corpses strewn across the wilderness”—these figures form the harrowing backdrop to dynastic decline. Yet the living, breathing persons behind these numbers—their cognitions, their choices, how they sought meaning amid despair, how they unconsciously participated in or resisted systemic evolution—are all erased. The old paradigm has no need to know these things, because its unit of analysis is the “dynasty” or “civilization”; individuals are merely the replaceable atoms that constitute these grand entities.

At the micro-explanatory level, individual behavior is reduced to passive response to systemic incentives. The old paradigm harbors an unexamined behavioral assumption: the individual is an input-output unit of systemic functions. Lighten taxes and corvée, and the people will flock to you; impose harsh laws and severe punishments, and the people will fear you; natural disasters and man-made calamities strike, and the people will rebel. This assumption reduces the individual to a “stimulus-response” machine devoid of internal cognitive structure, wholly ignoring what Axiom 4 reveals: **all cognition is constrained by the framework or lens through which it operates; the same individual, in the same objective situation, may produce radically different interpretations and actions depending on the internalized framework.**

More importantly, the old paradigm is entirely incapable of handling the **differentiation, leap, and convergence** of individual cognition. Within the dynastic cycle narrative, transformative forces in history are typically classified into two types: top-down reform by “wise rulers and virtuous ministers,” or bottom-up rebellion when “official oppression drives the people to revolt.” The former presupposes that individuals must be positioned at the center of power to exert influence; the latter presupposes that individuals will act collectively only when the baseline of survival is breached. Neither can explain a phenomenon that recurs throughout history yet has never been theorized: **at the system’s periphery, certain individuals invariably complete a cognitive framework leap first. They see what their contemporaries cannot see; they understand the world in new ways and begin to act; and their actions eventually converge into a force that changes the system.**

Copernicus and Galileo in the history of science, Luther and Calvin in the history of religion, Locke and Paine in the history of politics—these figures were neither power-holders in the “wise ruler and virtuous minister” mold nor famine-stricken rebels in the “oppression breeds revolt” mold. They were **cognitive probes** (Theorem B2)—awakened individuals who completed cognitive escape (Theorem D23) at the system’s periphery. The old paradigm has no theoretical place for such figures, because its analytical framework contains no dynamical possibility that “individual cognition can transcend the systemic framework.”

The consequences of this blind spot are profound. If the individual is merely the passive output of systemic functions, then individual rational action can only consist in optimization within the system’s rules—becoming a more efficient cog. If the individual can awaken, can escape, can construct new cognitive frameworks at the periphery, then history is not merely the cycle of dynasties but the accumulation and convergence of countless individual cognitive breakthroughs. The old paradigm chose the former, and thus could only see the cycle; the new paradigm chooses the latter, and thus perceives the possibility of phase transition.

[Cross-Corroboration through Chinese and Western Cases]

- **Chinese Case: Intellectual Heterodoxy in the Late Ming and Early Qing.** Huang Zongxi's fundamental critique of monarchy in *Waiting for the Dawn*—"the greatest harm in all under Heaven is the ruler alone"—far exceeded the expectations of his contemporaries, who merely hoped for "wise rulers and virtuous ministers." Gu Yanwu's distinction between "the fall of a state" and "the fall of all under Heaven" detached cultural identity from dynastic allegiance. These intellectual breakthroughs occurred at the systemic periphery of the collapsing Ming dynasty (among the remnant loyalist community). They could not save the Ming, yet centuries later they became part of the noospheric gene reserve that nourished modern intellectual transformation. The old paradigm saw only that the Ming fell to rebels and Manchus; it could not see the long-term seeds planted by these individual cognitive escapes.
- **Western Case: The "Invisible College" in the Early Modern Scientific Revolution.** The scientific revolution of the seventeenth century was not led by universities or royal courts—universities of the time remained firmly locked within the scholastic paradigm. The true breakthroughs occurred at the system's periphery: a dispersed network of scholars across Europe (Galileo, Kepler, Descartes, Boyle, etc.) exchanged observational data and theoretical conjectures through private correspondence, forming an informal "Invisible College." Each of these individuals had completed cognitive escape from the Aristotelian physical paradigm, and through their epistolary network, their insights converged into a new cognitive framework. When Newton synthesized their achievements into the system of classical mechanics, the critical threshold of the scientific revolution's phase transition was breached. The old paradigm can explain this as the "spirit of the age" or the "chance emergence of genius," but it cannot reveal how the **synergetic cognitive network** among these individuals, beginning from the periphery, gradually displaced the old system of knowledge production.

2.3 The Third Blind Spot: The Dynamically Unexamined Illusion of Reform

The most insidious and enduring legacy of the old paradigm is its implicit indulgence in the possibility of "moderate reform." This indulgence does not appear as an explicit proposition—indeed, if the cycle were truly an iron law, reform would be logically excluded. Yet at the margins of the cyclical narrative, an unspoken hope persists: if the problem lies in institutional decay and ruler incompetence, then perhaps through institutional adjustment and moral exhortation, decline can be delayed or even prevented *without* overthrowing the entire system.

It is within this logical crevice that reformers across the ages have launched their endeavors. They do not seek to overthrow the system but to repair it from within. They employ the tools the system provides—imperial edicts, bureaucratic procedures, legitimizing discourses—to combat the system's maladies. They believe that with sound methods and resolute will, the system can renew itself.

The verdict of history is this: this belief has never been verified, yet it has been repeatedly falsified.

Wang Anshi's reforms, backed by the imperial authority of Emperor Shenzong of Song, were launched with the defiant spirit of "fear not the alterations of Heaven, follow not blindly the ancestors, heed not the gossip of men." What was the outcome? The Green Sprouts Policy (*Qingmiao fa*), intended to suppress usury and relieve peasant distress, was distorted in implementation by local clerks into an instrument of coercive extraction. The Hiring Service Exemption Policy (*Mianyi fa*), meant to equalize corvée burdens, became a new avenue for bureaucratic rent-seeking. The reforms touched the parasitic interests of the great bureaucratic landlords and great merchants, and these interest groups, leveraging

the bureaucratic networks in which they were deeply embedded, systematically distorted implementation and mobilized public opinion against the reforms. When Shenzong died, the conservative faction returned to power and the new policies were entirely abolished. **The tools the reformer used (imperial edicts, the bureaucratic apparatus) were precisely the same tools upon which the systemic parasites depended for their survival and through which they captured and reversed the reforms.**

Zhang Juzheng's reforms employed the Performance Evaluation Method (*Kaocheng fa*) to discipline the bureaucracy and the Single Whip Method (*Yitiao bianfa*) to simplify taxation, producing a brief period of resurgence in the early Wanli era. Yet the fruits of reform were entirely dependent upon Zhang Juzheng's personal authority and the Wanli Emperor's forbearance. When Zhang died, the anti-reform bureaucratic faction immediately launched a comprehensive counterattack. The Performance Evaluation Method was abandoned; the Single Whip Method gradually deviated in implementation; the Ming dynasty's fiscal and administrative systems swiftly reverted to their old trajectories. **"When the man perishes, his policies perish"—this is not a lament over Zhang Juzheng's personal fate, but a precise description of the dynamical constraint that intra-systemic reform inevitably depends upon personal authority, and personal authority cannot be institutionally transmitted.**

In the late Roman Republic, the Gracchi brothers attempted to use legal procedures—the tribunician legislative power, the plebeian assembly vote—to redistribute the public land monopolized by the aristocracy. They scrupulously observed every legal tool the republican constitution provided. What was the outcome? Tiberius Gracchus was lynched by the senatorial aristocracy on the charge of "aspiring to kingship." Gaius Gracchus, after pushing through more extensive reforms, was suppressed by military force under the Senate's "Ultimate Decree" and took his own life. **Legal tools, when they touched fundamental interests, were violently destroyed by the interest groups in the name of "defending the Republic." The tools could not protect the tool-user, because the boundaries within which the tools could be used were precisely demarcated by the legitimacy that the parasites themselves defined.**

Gorbachev's reforms sought to revitalize the system and correct its accumulated maladies within the framework of the Soviet Communist Party, through *glasnost* (openness) and *perestroika* (restructuring). The tools he used were the authority of the CPSU General Secretary and the reformist consensus within the Party. What was the outcome? The reforms released decades of pent-up discontent, but the old system lacked institutionalized channels for processing this discontent. The parasite network accelerated its conversion of power into private wealth during the uncertainty of reform. Ultimately, the system was not repaired by reform but collapsed from the positive-feedback destabilization that reform triggered. **The reformer attempted to use the tools of the old system to remake the old system, and the tools disintegrated along with the system itself.**

This universal failure, spanning civilizations and epochs, cannot be explained away as "contingent errors" or "unripe conditions." It points toward a dynamical proposition that must be taken with the utmost seriousness: Are the endogenous tools of intra-systemic reform logically incapable of being used to rewrite the system's fundamental rules?

The old paradigm has never posed this question. It tacitly assumes that the tools of reform are "neutral" and that success depends on the user. But tools are never neutral. Tools are the embodiment of systemic rules. The imperial edict is the embodiment of pyramidal structure; bureaucratic procedure is the embodiment of hierarchical command; the tribunician office is the embodiment of the republican constitution; intra-Party democratic centralism is the embodiment of the Leninist party. **When you use**

these tools, you have already accepted the rule-boundaries that these tools presuppose. You can optimize within those boundaries, but you cannot use those tools to dismantle the boundaries themselves. To attempt to do so is like trying to lift yourself off the ground by your own hair—logically impossible, dynamically doomed.

The exposure of this blind spot drives us toward an inescapable conclusion: **Moderate reform, as an intra-systemic strategy, has its effective domain strictly confined within the “safe zone” tolerated by the systemic parasites.** Any reform that touches the fundamental interests of the parasites, regardless of how legal its means or how resolute its will, will trigger the system’s immune rejection response—either being distorted and absorbed, violently terminated, or perishing along with the system itself. Reform is not a middle path; it is a dynamical illusion.

[Cross-Corroboration through Chinese and Western Cases—Supplementary Cases]

- **Chinese Case: The Self-Strengthening Movement and the Hundred Days’ Reform of the Late Qing.** The Self-Strengthening Movement attempted to achieve “self-strengthening” by introducing Western technology without altering the imperial framework. The tools it used were reformist officials within the old bureaucracy (Zeng Guofan, Li Hongzhang, etc.) and limited imperial authorization. The reforms achieved localized technical successes (modern arsenals, shipping, telegraphy), but halted before touching the political system. The Hundred Days’ Reform sought to go further, pursuing institutional reform (abolishing the eight-legged essay, establishing a parliament, cutting redundant offices), and was violently terminated within a mere 103 days by the parasite network represented by Empress Dowager Cixi. The two cases together demonstrate: **Technical reform can occur within the safe zone tolerated by the system; institutional reform, the moment it touches fundamental rules, immediately collides with the reform boundary.**
- **Western Case: Fiscal Reform Efforts before the French Revolution.** Successive finance ministers under Louis XVI—Turgot, Necker, Calonne—attempted to rescue the bankrupt Old Regime through reform measures including tax increases, reduction of privileges, and fiscal unification. The tools they used were royal authority and traditional administrative procedures. Every reform attempt was defeated by the combined resistance of the *parlements* (controlled by the nobility of the robe) and the privileged orders. Ultimately, Louis XVI, having exhausted all options, was forced to convene the Estates-General—an act that itself had already crossed the boundary of “intra-systemic reform” and entered the trajectory of “systemic reconstruction.” The Old Regime was not overthrown by revolution; it opened the door to reconstruction itself after repeated failures of reform attempts collided with the boundary. **This process perfectly illustrates the dynamics of the reform boundary: when reform exhausts all legal tools and still cannot break through, the system either collapses or undergoes phase transition.**

2.4 The Internal Nexus of the Threefold Blindness and Their Collective Consequence

These three blind spots are not isolated; they constitute a mutually reinforcing logical closed loop:

- **Superficiality of attribution** prevents the identification of systemic parasites as the true dynamical etiology, thereby deflecting attention toward surface problems that seem “fixable” (slack enforcement, poor appointments);
- **Erasure of the individual** prevents the imagination of any third path of transformation beyond “wise rulers and virtuous ministers” and “oppression breeds revolt,” thereby pinning

hopes on elite reform within the system;

- **The illusion of reform**, nourished by this cognitive vacuum, acquires unexamined legitimacy, driving generation after generation of reformers into endeavors doomed to fail, and each failure in turn reinforces the fatalistic belief that “the cycle of rise and fall is irresistible.”

The ultimate product of this closed loop is the core illusion that the narrative of the dynastic cycle sustains: **History is cyclical, the individual is powerless, and effort is ultimately futile.**

The remainder of this paper will dismantle each node of this loop in turn. We will demonstrate:

Attribution can be precise—the proliferation of systemic parasites is an identifiable, measurable dynamical process. The individual can be active—cognitive escape is a real dynamical possibility, and awakened individuals can converge into a collective force capable of rewriting the rules. The boundary of reform can be proven—it is not an empirical generalization but a dynamical theorem deducible from systemic axioms.

Having dismantled the old paradigm, we shall proceed to lay the foundations of the new.

Part Three: The Axiomatic Foundation of the New Paradigm

Critique of the old paradigm is merely the first step. If this paper were to stop at exposing the blind spots of the dynastic cycle, it would be but one more addition to the critical literature. A paradigm displaces another not by demonstrating the latter’s defects, but by furnishing an alternative framework that is manifestly superior in both explanatory and predictive power.

The task of this section is to lay the axiomatic foundation for just such an alternative. I shall present, in their most compact form, the five core axioms drawn from *Noospheric Ecology and Civilization Dynamics* that bear directly upon the problem of the civilizational cycle. Each axiom is given in its essential formulation: statement, exposition, and direct implication for civilizational questions. For more complete logical derivations and formalized expressions, the reader is referred to the original monographs.

Before presenting the axioms, it is necessary to declare a fundamental feature of the new paradigm: **reflexivity**. This paradigm acknowledges that it, too, is subject to Axiom 4 (the relativity of cognitive frameworks). It is not a closed dogma claiming possession of absolute truth. However, by taking the “relativity of cognitive frameworks” as its primary object of study and by internalizing Axiom 1 (logical self-reference) as the core operational mechanism of the theory, this paradigm acquires a unique **reflexive privilege**—it is an open cognitive operating system capable of observing, revising, and iterating upon itself. This means that every claim made in this paper, including the demonstration of the reform boundary and the prediction of phase transition conditions, is open to empirical test and possible revision. Reflexivity is not a weakness; it is the very foundation that distinguishes this paradigm from all closed ideologies.

3.1 Axiom 1: The Dynamical Principle of Logical Self-Reference

Statement:

In any noospheric system that has attained a certain threshold of complexity, its internal information processing will give rise to the capacity for self-reference, thereby forming a self-iterating “logical sub-universe.” Creative intuition, conscious reflection, and paradigm revolution are all manifestations of this principle at different scales.

Exposition:

“Self-reference” is the capacity of a system to take itself as an object of observation and operation. A stone cannot self-refer; a simple feedback loop (such as a thermostat) possesses only the most rudimentary form of self-reference. But when the complexity of information processing surpasses a certain threshold, a “model of the system itself” emerges within the system—it can examine its own operational state, evaluate its own cognitive frameworks, and attempt to revise them. The metacognitive capacity of humans (thinking about thinking) and the self-critical traditions of civilizations (reflection upon their own narratives) are both manifestations of this principle at the individual and collective scales.

Direct Implication for the Problem of the Civilizational Cycle:

If civilization were merely an accumulation of matter and institutions, it could not possess the capacity for “self-awareness,” let alone “break out” of any cycle. Axiom 1 reveals that **when a civilizational system attains a certain complexity, it engenders the capacity to observe its own operational state**. The carriers of this capacity are not “civilization” in the abstract, but those individuals within the system who have first completed the cognitive leap (second-order noospheric humans). The reason the dynastic cycle has never been “broken” is not that breaking it is logically impossible, but that old systems lack the institutionalized channels for converging individual awakenings into collective self-reference. One of the missions of the new paradigm is precisely to design such channels (the metacognitive interface).

3.2 Axiom 2: Material/Noospheric Duality**Statement:**

Any system that has attained a certain threshold of complexity presents itself simultaneously as a dual entity: a material substance and a noospheric pattern. The material state constitutes the system’s substrate and energetic foundation; the noospheric state is the pattern, rules, and representations of information organization both within the system and between systems. The two are mutually implicative and inseparable.

Exposition:

A human being is both a carbon-based physical body occupying space (material state) and a cognitive pattern composed of memory, belief, personality, and knowledge (noospheric state). When the body perishes, the noospheric state dissipates—unless it has been externalized and transmitted in symbolic form (writings, reputation, images). A civilization is both the sum of population, territory, cities, and armies (material state) and a network of cognitive protocols constituted by language, law, religion, and historical narrative (noospheric state). The key to understanding Axiom 2 is this: **the material state provides the substrate and energy, but what the system *is*—its structure, functions, goals, and meaning—is entirely defined by the noospheric state**.

Direct Implication for the Problem of the Civilizational Cycle:

The old paradigm’s explanations of civilizational rise and fall concentrate almost exclusively at the material level: demographic shifts, land concentration, fiscal surpluses and deficits, military strength. Axiom 2 reveals the fundamental limitation of this perspective: **the decay of the material state is typically the consequence, not the cause, of pathology in the noospheric state**. The Qing dynasty’s defeat in the First Sino-Japanese War can be explained at the material level by inferior ship speed and slower artillery fire. But the deeper noospheric explanation is this: the organizational paradigm of the navy (rooted in feudal private-army traditions), the military cognitive framework (dominated by a

continental strategic mindset), and institutional protocols (factional infighting, logistical corruption) had long since rotted away. Material disparity was the symptom; noospheric decay was the etiology. Without diagnosing the noospheric state, one cannot understand how the material state came to such a pass.

3.3 Axiom 3: The Self-Organization Tendency of Systems (Generalized Will-to-Exist)

Statement:

From galaxies to living organisms to noospheric systems, systems exhibit an intrinsic tendency toward self-organization that drives them away from thermodynamic equilibrium. This tendency is a universal phenomenon of local entropy reduction in the cosmos. At the noospheric level, this tendency manifests as the drive to maintain and expand one's own low-entropy ordered structure (i.e., one's noospheric state), and can be experienced as the "will-to-exist."

Exposition:

Axiom 3 furnishes the first mover of the new paradigm's dynamics. It answers a foundational question: Why does a system "seek to maintain itself"? Physics tells us that isolated systems spontaneously tend toward entropy increase (disorder). Yet life and civilization exhibit the opposite tendency—they actively draw energy and information from their environment to maintain and even enhance their own complexity. This "negentropic" inclination is not a mysterious vital force, but the structural stability that open systems spontaneously develop within energy flows. At the level of the human individual, it is experienced as the instinct for survival, the craving for meaning, the impulse to resist being forgotten or effaced. At the civilizational level, it manifests as the self-perpetuation of culture, the inertia of institutions, and the intergenerational transmission of narratives.

Crucial Supplement: The Dynamism of the Unit of Operation. The unit of operation of the will-to-exist—the boundary of the "self"—is not fixed to the biological individual but can expand to encompass any noospheric entity that the individual has internalized: family, lineage, nation, faith, ideology. A person who sacrifices for family honor, martyrs for religious faith, or dies for a political ideal—such acts are "irrational" at the biological level, but at the level of noospheric dynamics, they are the result of the will-to-exist operating through a unit that has expanded from the "biological individual" to the "internalized noospheric entity." **The individual gives their life precisely because this tendency is computing and driving action with a larger noospheric entity as its unit.**

Direct Implication for the Problem of the Civilizational Cycle:

This axiom leads directly to two core concepts. First, **the dynamical origin of systemic parasites:** once a subsystem is formed, it engenders its own self-organization tendency (will-to-exist). When the will-to-exist of the subsystem conflicts with that of the host system, the subsystem will prioritize its own maintenance—this is the deep mechanism behind the inevitable generation of systemic parasites. Second, **the dynamical resistance to reform:** the old system as a whole possesses a powerful self-organizational inertia. Any reform that touches the system's fundamental rules will trigger the immune rejection of the system (and especially of its parasitic subsystems). This is not a matter of morality but of dynamical necessity.

3.4 Axiom 4: The Relativity and Path-Dependence of Cognitive Frameworks

Statement:

There exists no absolutely objective thinking; all cognition is constrained by the "framework" or "lens" within which it operates. This framework not only filters information but actively constructs the very

“reality” that the observer perceives. An individual’s cognitive output is the integrated result of all past experiences, knowledge paradigms, and affective conditioning; cognitive structures possess powerful inertia.

Exposition:

Axiom 4 is the epistemological core of the new paradigm. It does not assert that “objective reality does not exist” or that “everything is subjective,” but rather reveals that **every path by which we access reality is mediated by the encoding and decoding of cognitive frameworks**. The same objective datum—say, a decline in a dynasty’s fiscal revenues—may be interpreted in the old framework as “the emperor is extravagant and officials are corrupt,” while in the new framework it may be interpreted as “the systemic parasite burden has surpassed the carrying capacity of the tax base.” Both interpretations “see” the same data, but the causal structures they “perceive” are radically different. More importantly, individuals are typically unaware that they are operating within a framework—the framework runs in the “background,” presenting its outputs as “common sense.” This is cognitive path-dependence: once a framework is deeply internalized, the individual tends to interpret all new information through this framework, even distorting information to fit the framework, thereby creating a self-reinforcing cognitive closed loop.

Direct Implication for the Problem of the Civilizational Cycle:

Axiom 4 provides the micro-cognitive foundation for understanding civilizational ossification. A civilization’s dominant narrative (e.g., “the ancestral statutes cannot be changed,” “free markets are automatically optimal”), once internalized as the default framework by the majority of society’s members, generates a powerful **noospheric gravity field**. Within this field, any information, viewpoint, or action that conflicts with the dominant framework is automatically filtered, rejected, or marginalized. This is the dynamic described by Theorem D23 (Cognitive Escape): for an individual to break through the dominant framework, they must expend enormous cognitive energy (metacognitive capacity) to counter the gravitational field constituted by the entire social consensus. Most individuals cannot afford this cost, and thus choose to optimize within the framework rather than break through it—this is the individual-level root of civilizational ossification. Axiom 4 also explains why “wise rulers” have difficulty identifying “treacherous ministers”: the ruler’s information-input channels (memorials, court deliberations, eunuchs, close advisors) have already been deeply encoded by the parasite network; the “reality” the ruler perceives is precisely the reality the parasite network wishes him to perceive.

3.5 Axiom 5: Symbol as Noospheric Gene

Statement:

Within the noospheric ecology, symbols such as concepts and narratives hold a status and function equivalent to biological genes. They are the fundamental units of cultural inheritance, variation, and selection.

Exposition:

Biological evolution depends upon the replication (inheritance), mutation (variation), and environmental screening (selection) of genes. Noospheric evolution depends upon the transmission (education, imitation), innovation and misreading (variation), and screening by social consensus and power structures (selection) of symbols. A concept—for instance, “freedom,” “equality,” “benevolence,” “righteousness,” “involution”—once widely accepted by a society, behaves like a successful gene, utilizing individuals’ cognitive resources to replicate itself (being repeatedly cited,

disseminated, internalized), and in the process competing with other symbols for attention and legitimacy. Symbols are not passive labels for reality; they are “meaning-organs” that actively shape reality. When the symbol “involution” (*neijuan*) was coined and popularized, countless individuals underwent a fundamental shift in their interpretation of their own life conditions—they “saw” a predicament they had previously lacked the language to articulate, and adjusted their behavior accordingly (withdrawing from excessive competition, seeking work-life balance). A single symbol altered the niche perception and action strategies of an entire generation.

Direct Implication for the Problem of the Civilizational Cycle:

Axiom 5 furnishes the operational mechanism for understanding civilizational transformation. If symbols are noospheric genes, then **civilizational phase transition is, in essence, a large-scale replacement of noospheric genes**—the old core narrative (e.g., “divine right of kings”) is displaced by a new core narrative (e.g., “popular sovereignty”). The difficulty of reform lies not in the resistance of “interest groups” (that is mere surface), but in the fact that the old symbolic system has already deeply encoded the cognitive protocols, legitimizing discourses, and meaning-systems of the entire society. If the reformer continues to use the old symbolic system (as when Wang Anshi cited the *Rites of Zhou* to justify his reforms), he is already fighting within the semantic field of the enemy—he may exploit certain elastic spaces within the old symbols, but he cannot fundamentally rewrite the rule-boundaries that the old symbols define. True paradigm revolution is necessarily accompanied by the invention of new symbols or the radical redefinition of old ones (the operation of the noospheric weak force). One key to the success of Shang Yang’s reforms was the use of the new symbol “military merit” to redefine the criteria of social status, thereby circumventing the old symbol-system of “bloodline” monopolized by the hereditary aristocracy.

3.6 Summary of the Axiomatic Foundation: From Five Axioms to Three Core Propositions

The five axioms presented above are not isolated dogmas but constitute a logically coherent dynamical framework:

- **Axiom 1 (Logical Self-Reference)** provides the capacity-basis for civilizational self-awareness.
- **Axiom 2 (Duality)** directs diagnosis to attend simultaneously to material and noospheric states.
- **Axiom 3 (Self-Organization Tendency)** reveals the dynamical roots of systemic parasite generation and resistance to reform.
- **Axiom 4 (Framework Relativity)** explains the cognitive mechanisms of civilizational ossification and the difficulty of individual escape.
- **Axiom 5 (Symbol as Noospheric Gene)** describes the operational realization of civilizational phase transition—the replacement and redefinition of symbolic systems.

From these five axioms, the core analytical tools of this paper can be derived, including but not limited to:

- **The Systemic Parasite Identification Theorem (Theorem D21):** Derived from Axiom 3 and Axiom 6 (the axiological criterion of systems), used to identify the dynamical lesion of civilizational decline.
- **The System Builder’s Advantage Theorem (Theorem D17):** Derived from Axiom 3 and Axiom 4, used to explain why participants cannot surpass the builder from within the system.
- **The Cognitive Escape Theorem (Theorem D23):** Derived from Axiom 1 and Axiom 4, used

to quantify the conditions under which an individual breaks through the dominant framework.

- **The System Topology Theorem (Theorem A0):** A synthesis derived from all axioms, providing the meta-framework for civilizational architecture design.

These theorems will be invoked and explicated as needed in the subsequent sections of this paper. For the present, the reader need only grasp this: **Every claim advanced in this paper is not an isolated insight or rhetorical flourish, but a corollary rigorously deduced from a logically self-consistent axiomatic system.** This is precisely what fundamentally distinguishes the new paradigm from the old—the old paradigm is an inductive summary of historical experience; the new paradigm is the logical unfolding of an axiomatic system. The former can only tell you “what happened in the past”; the latter can tell you “what may happen in the future, and what you can do about it.”

Having established the axiomatic foundation, the next section enters the core constructive phase of the new paradigm: proposing a new diagnostic criterion for civilizational health and demonstrating its comprehensive superiority over the old metrics in both explanatory and diagnostic power.

Part Four: The Core Criterion—Redefining the Health and Decline of Civilization

If the old paradigm’s attributions are superficial, then the new paradigm must furnish a criterion that is deeper in the causal chain—one that can explain not merely *when* civilization declines, but *why* it declines. The task of this section is to propose precisely such a criterion and to demonstrate its comprehensive superiority over traditional metrics (such as GDP, territorial expanse, military strength, and dynastic longevity) in both explanatory and diagnostic power.

4.1 The Formulation of the Civilization Potential Equation

In *Civilization Dynamics*, the health of a civilizational system is defined as a function constituted by the product of three core factors:

$$\text{Civilization Potential} = \Sigma(\text{Individual Niche Actualization Degree}) \times \text{Systemic Resource Fluidity} \times (1 - \text{Systemic Parasite Burden})$$

This formula is not an inductive fit to existing historical experience but a state function rigorously deduced from the axiomatic system. Its multiplicative form carries profound dynamical implications: **should any single factor approach zero, the entire Civilization Potential undergoes a nonlinear collapse.** This explains the most perplexing phenomenon in civilizational decline—how an apparently vast and wealthy empire can disintegrate with astonishing rapidity.

The following subsections explicate the dynamical meaning of each factor in turn.

4.2 The First Factor: $\Sigma(\text{Individual Niche Actualization Degree})$

Definition: The Individual Niche Actualization Degree measures the extent to which an individual realizes their potential within their given noospheric niche. A person who could have become a master artisan but is assigned by the system to repetitive labor may have an actualization degree of less than 30 percent; a person whose talents are fully engaged and who continuously deepens their craft in a fitting domain may have an actualization degree exceeding 80 percent. $\Sigma(\text{Individual Niche}$

Actualization Degree) is the sum of the actualization degrees of all individuals within the system.

Dynamical Meaning: This factor operationalizes Axiom 6 (the axiological criterion of systems in its humanistic formulation) into a measurable civilizational metric. It fundamentally inverts the evaluative logic of the old paradigm: **the strength of a civilization lies not in how many individuals it contains, but in how many individuals it enables to live out their innate gifts.** A system composed of a hundred million standard replicants may have a far lower Σ -value than a system composed of ten million second-order noospheric humans—because each individual in the latter contributes unique cognitive output to the system from their optimal niche.

Historical Corroboration: Athens at its Periclean zenith had a total population of merely three hundred thousand (including slaves and resident aliens), with adult male citizens numbering only about forty thousand. Yet this minuscule system engendered creative achievements—in philosophy, drama, sculpture, history, mathematics, and architecture—that remain foundational to human civilization. Its Σ -value was extraordinarily high. By contrast, the Persian Empire possessed millions of subjects and vast material wealth, but individual niches were rigidly locked into the fixed gears of the imperial machine; its Σ -value was exceedingly low. The outcome of the Greco-Persian Wars, viewed at the material level, appears a miracle of “the small defeating the large”; viewed at the noospheric level, it was the inevitable triumph of a high- Σ system over a low- Σ system.

Contrast with Old Metrics: The old paradigm’s metrics of civilizational strength—population, GDP, territory—all remain at the level of “how many individuals.” The new criterion asks “how alive are the individuals.” A society with rising GDP but suffocating individuals is a “flourishing age” in the old paradigm; in the new paradigm, it is a pathological state.

4.3 The Second Factor: Systemic Resource Fluidity

Definition: Systemic Resource Fluidity measures the ease with which information, opportunity, talent, and capital flow across hierarchical levels and domains within the system. Its value ranges from 0 (completely obstructed) to 1 (completely unimpeded).

Dynamical Meaning: The essence of resource fluidity is the health of the system’s topological structure. A healthy civilization must maintain a dynamic equilibrium between pyramidal structure (which handles deterministic tasks and secures survival) and distributed structure (which handles uncertainty and drives innovation) (Theorem A0). Resource fluidity is precisely the key indicator of whether an effective **constitutional interface** exists between these two structures. When fluidity is high, innovations from the bottom can rapidly disseminate through the system and receive resource support; cognitive probes at the periphery can be recognized and integrated by the system; parasites find it difficult to entrench themselves by monopolizing information channels. When fluidity approaches zero, the system fractures into mutually impermeable, rigidified strata—the upper strata block the ascent of the lower, and the lower cannot transmit information upward. At this point, civilization has entered a state of **cognitive deadlock**.

Historical Corroboration: The Tang dynasty at its height owed much of its vitality to the early establishment of the civil service examination system (*keju*) during the Sui and early Tang, which opened an institutionalized channel of upward mobility for the lower strata of society. Though narrow in absolute numbers, this channel symbolically proclaimed that “bloodline is not the sole criterion,” thereby dramatically enhancing the entire system’s resource fluidity and cognitive vigor. The late Qing, by contrast, exhibited the opposite: the examination system had functionally degenerated into formulaic eight-legged essays, selecting no longer for talent but for conformity. Simultaneously, the widespread

practice of purchasing official posts (*juanguan*) turned the channel of mobility into outright venal transaction; the parasite network had completed its capture of the channels of flow. Fluidity approached zero; the system entered deadlock.

Contrast with Old Metrics: The old paradigm asks whether society is “stable”; the new criterion asks whether society is “fluid.” A hyper-stable system maintained by rigid order (such as late imperial China or the late Soviet Union) may have fluidity approaching zero. Such stability is not health but the precursor to ossification.

4.4 The Third Factor: (1 – Systemic Parasite Burden)

Definition: The Systemic Parasite Burden is a ratio between 0 and 1: **the total value extracted and dissipated by systemic parasites / the total value created by the system.** Systemic parasites, as defined by Theorem D21, are subsystems that maximize their own value extraction by systematically distorting rules and evading their due responsibility for maintaining the host system. (1 – Systemic Parasite Burden) is thus the proportion of net resources actually available for the system’s self-maintenance and development.

Dynamical Meaning: This factor is the most diagnostically potent component of the new criterion. It directly reveals the dynamical root of civilizational decline: **it is not that the system’s capacity to create value declines, but that the proportion of resources appropriated by non-value-creating parasites within the system relentlessly rises.** As a civilization enters its mature phase, its institutional structures grow increasingly complex, and various subsystems (bureaucratic agencies, privileged strata, monopolistic industries) begin to prioritize their own survival and expansion over the holistic interests of the host system. This process is dynamically inevitable—Axiom 3 (self-organization tendency) dictates that any subsystem, once formed, engenders a will-to-exist that operates for its own maintenance. In the absence of effective immune mechanisms, the parasite burden rate must monotonically increase over time.

Historical Corroboration: By the late Ming dynasty, the imperial clan had proliferated to several hundred thousand members. According to ancestral statutes, these clansmen were forbidden from engaging in any of the traditional four occupations (scholar, farmer, artisan, merchant) and were supported entirely by the state treasury. This item alone annually consumed more grain than the total land tax revenue of the entire empire. Concurrently, bureaucratic corruption, the collapse of the military farm system (*juntun*), and the intensification of land concentration drove the parasite burden rate far beyond the system’s carrying capacity. The Chongzhen Emperor reigned for seventeen years, assiduously attending to state affairs, practicing frugality and diligence—yet the Ming dynasty fell nonetheless. The old paradigm cannot explain this tragedy; it can only sigh that “the ruler was no ruler of a doomed dynasty, yet all his ministers were ministers of doom.” The new criterion furnishes a precise diagnosis: **when the parasite burden rate exceeds a critical threshold (approximately 30%), regardless of the ruler’s personal qualities, the system has already lost the resource elasticity necessary to absorb any internal or external shock.** Chongzhen’s diligence was futile optimization within a dying system whose parasite burden had long since passed the critical threshold.

Contrast with Old Metrics: The old paradigm measures fiscal health by “annual revenue” and “surplus.” The new criterion asks: how much of that revenue is swallowed by parasites? A system with enormous revenue but an even higher parasite burden rate is dynamically more fragile than a system with smaller revenue but a higher proportion of net resources. This explains why many “prosperous” dynasties proved so brittle in the face of crisis—their prosperity was a statistical illusion created after

parasitic extraction.

4.5 The Dynamical Implication of the Multiplicative Form: Explaining Nonlinear Collapse

The multiplicative form of the three factors possesses a crucial dynamical property: **the weakest-link effect is exponentially amplified.**

Consider a hypothetical system where individual actualization is decent (0.7), resource fluidity is moderate (0.6), but the parasite burden rate has reached 0.5 (half of all social wealth is consumed by parasites). The Civilization Potential is then $0.7 \times 0.6 \times 0.5 = 0.21$. The system continues to function, but it is already operating in a critical regime.

When an external shock (war, natural disaster) causes resource fluidity to plummet from 0.6 to 0.3, the Civilization Potential collapses to $0.7 \times 0.3 \times 0.5 = 0.105$. The system suddenly discovers that it cannot mobilize resources to meet the crisis—not because resources do not exist, but because resources are locked down by parasites and the channels of flow are obstructed. Collapse occurs with startling speed. Observers marvel that “so vast an empire could disintegrate in an instant,” unaware that the system had been running near the critical threshold for decades.

This is the explanation that Civilization Phase Transition Dynamics offers for “sudden collapse”: **the collapse is sudden, but the system’s entry into the critical regime is gradual and measurable.** The old paradigm can only retrospectively identify “triggers”; the new paradigm can, by monitoring the historical trends of the three factors, provide early warning of when the system is approaching criticality.

4.6 From Criterion to Pathology: Two Core Dynamical Mechanisms

The Civilization Potential formula is a diagnostic tool; it tells us “where the disease lies.” But to understand “where the disease comes from,” we must delve into two core dynamical mechanisms derived from the axiomatic system.

First Mechanism: The Inevitable Proliferation of Systemic Parasites (Theorem D21 and Theorem D22).

Any complex system, in the course of its evolution, generates functionally differentiated subsystems. At their inception, these subsystems all bear the original function of serving the holistic health of the host system (Axiom 7). However, once a subsystem acquires a stable resource input and a degree of rule-making autonomy, its internal will-to-exist (Axiom 3) will drive it to prioritize “maintaining and expanding itself” over “serving the host system.” This process is termed **subsystem functional alienation** (Theorem D22).

In the early stages of functional alienation, the subsystem can still maintain a balance between serving the host and self-expansion. But over time, the subsystem leverages its informational advantages and its participation in rule-making to systematically distort the host system’s rules in favor of its own extraction. When the degree of alienation surpasses a critical point, the subsystem transforms into a **systemic parasite** (Theorem D21)—its existence no longer promotes but persistently damages the holistic health of the host system.

Crucial Corollary: **The generation of systemic parasites is not a matter of contingent moral decay, but a dynamical inevitability of structural evolution in complex systems.** Any civilization that lacks institutionalized immune mechanisms against functional alienation (metacognitive interfaces, independent criteria for diagnosing alienation, procedures for excising parasites) will inevitably see its parasite burden rate monotonically increase over time. The countdown to civilizational decline begins

the moment the civilization is built.

Second Mechanism: The Systematic Obstruction of Pathways for Cognitive Escape (Theorem D23 and Theorem D24).

The proliferation of systemic parasites damages the material and institutional dimensions of civilization. The obstruction of cognitive escape pathways damages the noospheric dimension—it deprives the system of the cognitive capacity for self-correction.

As Axiom 4 demonstrates, the dominant cognitive framework generates a powerful noospheric gravity field. Within this field, for an individual to break through the framework and achieve cognitive escape, their metacognitive capacity M must be sufficient to counter the product of the paradigm’s authority A and the number of individuals internalizing it N (Theorem D23). During a civilization’s ascending phase, the dominant framework can still tolerate a degree of heterodox thinking; the cognitive fluidity constant k^* is relatively high, and the individual escape probability $P(\text{escape})$ is relatively large. But as civilization enters its ossification phase, the dominant framework, in order to preserve its own authority, systematically suppresses heterodox cognition—marginalizing, stigmatizing, or physically eliminating dissenters. Simultaneously, specialized education systems (Theorem B3) mold individuals into highly efficient “differentiated cells,” sharpening their cognition within a single framework while degrading their capacity to switch between frameworks. A and N continue to increase; M is continuously suppressed; $P(\text{escape})$ approaches zero. When the pathways for cognitive escape are completely obstructed, civilization loses the capacity to generate paradigm revolution from within. It can only await the next collapse as the parasite burden rate relentlessly climbs—this is the dynamical essence of the dynastic cycle.

4.7 The Universality of the New Criterion: The Same Regularity Across Chinese and Western Civilizational History

To demonstrate that the new criterion is not a particularistic explanation applicable only to a single civilization, this section presents two paired sets of Chinese and Western cases to exhibit its diagnostic power.

Case Set One: The Western Roman Empire and the Eastern Han Dynasty.

Criterion Factor	Western Roman Empire (3rd–5th centuries CE)	Eastern Han Dynasty (2nd–3rd centuries CE)
Σ(Individual Niche Actualization Degree)	After universal extension of citizenship, citizenship lost its original meaning of political participation; the urban middle class was crushed by inflation and heavy taxation; peasants were bound to large estates (<i>coloni</i>), becoming de facto serfs. Σ -value continuously declined.	The recommendation system (<i>chaju</i>) was monopolized by powerful gentry clans; the political criticism of Imperial University students was suppressed by the Disasters of Partisan Prohibitions; peasants, under land concentration, became tenants of powerful magnates or refugees. Σ -value continuously declined.
Systemic Resource	Senatorial aristocracy and equestrian order rigidified; the army transformed	Powerful gentry clans monopolized official careers and scholarship;

Criterion Factor	Western Roman Empire (3rd–5th centuries CE)	Eastern Han Dynasty (2nd–3rd centuries CE)
Fluidity	from citizen militia into barbarian mercenary forces, severing the connection between society and military; Diocletian’s reforms legally mandated hereditary occupations, institutionally locking fluidity.	eunuchs and consort clans alternately controlled central power channels; local magnates commanded private armed forces, and the center’s capacity to extract resources from localities collapsed. Fluidity approached zero.
Systemic Parasite Burden	The court bureaucracy swelled; the senatorial aristocracy held vast tax-exempt lands; military commanders wielded private armies, with military expenditure consuming the majority of fiscal revenue while providing ineffective defense. Parasite burden rate far exceeded the critical threshold.	The imperial clan population exploded, with stipends draining the treasury; eunuch and consort clan factions alternately seized power and wealth; local magnates concealed households and intercepted tax revenues. Parasite burden rate far exceeded the critical threshold.
Shared Diagnosis	The trajectories of decline across all three factors in both systems exhibit profound structural isomorphism. Neither was suddenly destroyed by “barbarian invasion” or the “Yellow Turban Rebellion”—these external shocks were merely the final straws that broke systems whose parasite burden rates had long since exceeded the critical threshold.	

Case Set Two: The Spanish Empire in the Age of Discovery and the “High Qing” Era.

Criterion Factor	Habsburg Spanish Empire (16th–17th centuries)	Qing Dynasty “High Qing” Era (18th century)
Σ(Individual Niche Actualization Degree)	The influx of American silver induced the “resource curse,” causing manufacturing atrophy and a social valorization of aristocratic disdain for commerce and industry; the Inquisition suppressed intellectual heterodoxy. Σ-value was extremely low.	Literary inquisitions persisted for over a century, silencing the scholarly stratum and forcing scholarship into philological exegesis, drying up original thought; population explosion intensified the man-land ratio, and involution reduced individuals to gears on the

Criterion Factor	Habsburg Spanish Empire (16th–17th centuries)	Qing Dynasty “High Qing” Era (18th century)
		subsistence line. Σ -value was extremely low.
Systemic Resource Fluidity	The statutes of purity of blood (<i>limpieza de sangre</i>) rigidly stratified society by lineage and religious background; the monopoly over American trade was controlled by the Consulado of Seville. Fluidity was institutionally locked.	The examination channel, though formally maintained, had degenerated into formulaic competition in the eight-legged essay; the sale of offices permitted direct conversion of wealth into power; fluidity was captured by parasites.
Systemic Parasite Burden	Court luxury, foreign wars (the Netherlands, Italy, the Armada) drained American silver; the nobility and Church held vast tax-exempt lands; the crown defaulted on its debt multiple times. Parasite burden rate was extremely high.	Routine corruption of the bureaucracy (the system of “deliberative fines” institutionalized graft); the parasitic Eight Banners stratum was entirely supported by the state; public works such as river conservancy and grain transport became abysses of graft. The parasite burden rate continued to climb beneath the surface of the “flourishing age.”
Shared Diagnosis	Beneath the surface of their respective “Golden Ages,” the new criterion reveals a startling fragility. The Spanish Empire declined rapidly in the seventeenth century; the Qing Dynasty’s underlying weakness was exposed by the White Lotus Rebellion at the end of the eighteenth century. A “flourishing age” is not proof of health but a phase in which the parasite burden rate can still be masked by imperial expansion or population growth. Once incremental resources are exhausted, accumulated contradictions erupt.	

These two case sets collectively demonstrate that **the criterion of Civilization Phase Transition Dynamics is universal**. Whether in the Mediterranean world or on the East Asian continent, whether in

classical empires or early modern dynasties, the dynamical pathways of decline exhibit striking structural isomorphism. The old paradigm treats the decline of each civilization as a unique historical event; the new paradigm reveals that they share the same underlying equations.

4.8 From Diagnosis to Action: The Practical Orientation of the New Criterion

The value of the new criterion lies not only in explaining the past but in guiding the future. It provides a roadmap for action that the old paradigm could never offer:

- **To enhance Civilization Potential, one need not pursue unlimited expansion of the material base**—that typically merely provides more objects for parasitic extraction. The first priority is **suppressing the systemic parasite burden rate**: identifying parasites, cutting off their resource supply, and establishing metacognitive interfaces to institutionally discover and excise parasites.
- **Second, enhance systemic resource fluidity**: dismantle stratum barriers, break information monopolies, and create institutionalized ascent channels (constitutional interfaces) for peripheral innovation and grassroots wisdom.
- **The ultimate aim is to raise Σ (Individual Niche Actualization Degree)**: to shift the civilizational systems of education, occupation, and evaluation from the old paradigm of “screening replaceable parts” to the new paradigm of “empowering each individual to attain their niche optimum.” This is the ultimate demand of Axiom 6 (the axiological criterion of systems in its humanistic formulation).

These three directions for action are entirely invisible within the old paradigm. The old paradigm can only counsel rulers to “lighten taxes and corvée, select the worthy and appoint the capable”—moral maxims that are eternally correct but eternally incapable of arresting decline. The new paradigm furnishes dynamical strategies that are actionable, measurable, and testable.

Having established the new criterion for civilizational health, the next section proceeds to the most crucial argument of this paper: **deducing from the criterion the dynamical boundary that renders “moderate reform impossible.”** We shall formally propose and prove **Corollary D17.1: The Boundary Theorem of Intra-Systemic Reform**, and test its existence through comparative analysis of reform cases across Chinese and Western history.

Part Five: The Dynamical Boundary of Reform—The Formal Proposal of Corollary D17.1

In Part Four, we proposed a new diagnostic criterion for civilizational health and revealed two core dynamical mechanisms: the inevitable proliferation of systemic parasites and the systematic obstruction of pathways for individual cognitive escape. Together, these two mechanisms explain why civilization moves from health toward ossification. Yet they have not yet directly addressed a question of more immediate practical urgency: **When civilization has already entered its ossification phase, can individuals or groups situated within the system possibly repair it through “moderate reform” and reverse the trajectory of decline?**

This question is urgent because it bears directly upon the action-choices of every individual who has

become aware of systemic pathology. If reform is possible, then the rational strategy is to accumulate strength within the system and pursue gradual correction. If reform confronts an impassable dynamical boundary, then continuing to pin hopes on reform is not prudence but complicity with the pathology—it will expend the energies of the awakened in repeated failures, delaying the arrival of genuine transformation.

The task of this section is to rigorously argue the latter case: **Moderate reform, as an intra-systemic strategy, has its effective domain strictly confined within the “safe zone” tolerated by the systemic parasites; any reform attempt that touches the fundamental rules of the system is dynamically doomed to failure.** This argument proceeds in four steps: first, restating Theorem D17 (System Builder’s Advantage) as the logical point of departure; second, deducing Corollary D17.1 (The Boundary Theorem of Intra-Systemic Reform) from Theorem D17, formally naming and stating it; third, empirically testing the corollary through comparative case analysis of reform efforts across Chinese and Western history; and fourth, preemptively addressing the potential objection of “reform initiated by the system builder,” thereby foreclosing the logical escape hatch of the reformist illusion.

5.1 Logical Point of Departure: Theorem D17—The System Builder’s Advantage Theorem

In *Noospheric Ecology*, Theorem D17 is stated as follows:

No participant in any system can defeat the system builder, because the builder possesses the ultimate power to define the direction in which the system’s dominant self-organization tendency expresses itself. All behaviors and tools of participants are embodiments of the system’s rules; the expression of their self-organization tendency is constrained within the direction set by the builder.

The dynamical foundation of this theorem lies in the combination of Axiom 3 (the self-organization tendency of systems) and Axiom 7 (the principle of functional derivation). A system—whether a state, a corporation, or an academic paradigm—once built and stabilized in operation, will engender a dominant direction of self-organization within it. This direction is not an abstract “will” but is concretely embodied in the fundamental rules of the system: who has the authority to make rules, who has the authority to interpret rules, who has the authority to enforce rules, how resources are allocated, and whence legitimacy derives. The sum of these fundamental rules constitutes the “buildership” of the system.

The system builder is that actor who is **capable of defining and modifying these fundamental rules.** They need not be the original founder of the system—the founder’s power may long since have been diluted by successors or usurpers—but they invariably occupy the unique position within the system that can legitimately rewrite the meta-rules. System participants are those actors who can only act within the given rules. Participants may optimize their strategies, may find spaces within the interstices of the rules, and may even become victors within the system through outstanding performance—but they **cannot alter the rules themselves**—because the power to alter the rules is precisely what the system they participate in refuses to grant them.

This asymmetry is the key to understanding all predicaments of intra-systemic reform.

5.2 Corollary D17.1: The Boundary Theorem of Intra-Systemic Reform

From Theorem D17, combined with Axiom 3 (the self-organization tendency of subsystems) and Theorem D22 (functional alienation of subsystems), we can deduce a corollary that directly targets the action of “reform.”

[Corollary D17.1: The Boundary Theorem of Intra-Systemic Reform]

Statement: In any system where the dominant direction of self-organization is defined by the system builder, any attempt at “moderate reform” that relies upon the system’s endogenous tools and legitimizing procedures will inevitably have its effects converge within the boundary of toleration set by the system builder (and the systemic parasites symbiotic with it). It cannot reach the rewriting of the system’s fundamental rules. Its long-term outcome will be either capture and distortion by systemic parasites, or forcible termination by the system’s immune mechanisms upon touching the boundary.

Criterion: The distinction between “reform” and “reconstruction” lies not in the radicalness of the means employed, but in **whether the action attempts to alter the direction of the system’s dominant self-organization without becoming a new system builder**. If the answer is affirmative, the action belongs dynamically to the category of reform and is subject to the constraint of this theorem.

Derivation:

1. **The tools of reform come from the system.** The defining characteristic of moderate reform is precisely its refusal to overthrow the existing system and its attempt to use the system’s own legitimate tools—edicts, procedures, discourses, powers of appointment—to correct the system’s maladies. Wang Anshi used the imperial edicts of Emperor Shenzong and the Finance Planning Commission; Zhang Juzheng used the Performance Evaluation Method and the Grand Secretariat’s powers of remonstrance; the Gracchi brothers used the tribunician legislative power and the plebeian assembly; Gorbachev used the authority of the CPSU General Secretary and intra-Party reform procedures. All these tools, without exception, are embodiments of the system’s rules.
2. **Tools presuppose boundaries of use.** No tool is neutral. The effectiveness of an imperial edict depends upon a highly centralized pyramidal bureaucratic structure; the effectiveness of the Performance Evaluation Method depends upon an administrative system centered on documentary assessment; the effectiveness of the tribunician institution depends upon the balance of powers within the Roman republican constitution; the effectiveness of intra-Party reform procedures depends upon the hierarchical obedience of democratic centralism. **When you use these tools, you have already tacitly accepted the systemic rule-framework within which these tools are embedded.** You may promote a certain degree of interest-redistribution within the framework, but you cannot use these tools to dismantle the framework itself—because the very design principle of the tools is to maintain the stability of the framework.
3. **Parasite control over the tools.** After the system has entered its ossification phase, the subsystems that execute the tools of reform (bureaucracies, assemblies, Party organs) have long since been deeply penetrated or directly controlled by systemic parasites. Reform directives, issued from the builder (or reformer) and transmitted through these subsystems, inevitably undergo **dynamical attenuation**: they are filtered layer by layer, selectively implemented, distorted in interpretation, and even reverse-utilized to strengthen the parasitic structure. The reformer’s will cannot act directly upon the object of reform; it must pass through the transmission medium controlled by parasites—and the medium itself is the source of resistance to reform.
4. **The system’s immune rejection.** When reform touches the fundamental interests of the parasites—that is, when it touches the redistribution of the system’s fundamental rules—the parasite network will activate the system’s immune rejection response. The forms of rejection

include: counter-attack through legitimizing discourse (“contrary to ancestral statutes,” “destroys republican traditions,” “betrays Party principles”), procedural obstruction (abuse of rules of deliberation, impeachment, veto), and, as a last resort, violent suppression. At this juncture, the reformer confronts a paradox: to break through the immune rejection, they must employ means that exceed the scope of the system’s legitimate tools—and by definition, they are no longer a “moderate reformer” but have become a “system reconstructor” (a revolutionary). If they insist on using only legitimate tools, they will inevitably fail in the face of immune rejection.

In summary, moderate reform is dynamically locked within a tragic contradiction: **the deeper the reform proceeds, the more it must use tools controlled by parasites; the more it uses these tools, the more the reform’s effects are distorted; the closer it approaches fundamental rules, the more intense the immune rejection it triggers.** This is a convergent process—however much initial energy the reformer invests, the ultimate effect necessarily converges within the safe boundary tolerated by the system.

5.3 Historical Case Verification: The Structural Isomorphism of Failed Reforms Across China and the West

Corollary D17.1 is a strong claim. It asserts that the failure of moderate reform is not contingent but dynamically inevitable. The way to test this claim is to examine reform cases across different civilizations, epochs, and institutional types: if their failures exhibit profound structural isomorphism, then the dynamical explanation gains robust empirical support.

The following analysis examines four cases in parallel—two from Chinese history, two from Western history.

Case One: Wang Anshi’s Reforms in the Northern Song Dynasty (1069–1085)

- **Reform Objectives:** Suppress land concentration, equalize taxation and corvée, increase fiscal revenue, strengthen national defense.
- **Tools Employed:** The imperial authority of Emperor Shenzong; the newly established Finance Planning Commission as the reform headquarters; a series of new policies (Green Sprouts Policy, Hiring Service Exemption Policy, Market Exchange Policy, Square-Field Land Tax Policy, etc.) as policy instruments.
- **Parasite Network:** A tripartite alliance of great bureaucrats, great landlords, and great merchants. They monopolized the power of local tax collection, market pricing, and the information channels to the court.
- **Process of Dynamical Attenuation:** The Green Sprouts Policy was originally designed as low-interest government loans to peasants to suppress usury. But in implementation, local clerks, to meet lending quotas, coercively imposed loans on well-off peasants who did not need them, and even converted the loans into a de facto supplementary tax. The Hiring Service Exemption Policy, meant to equalize corvée burdens, became a new category of bureaucratic fees. The Market Exchange Policy, meant to stabilize prices, evolved into official monopolization of markets at the expense of private commerce. **The tools of reform, in passing through the parasite-controlled bureaucratic network, were systematically distorted into instruments of reverse extraction.**
- **Immune Rejection:** The conservative faction, wielding the legitimizing discourse of

“ancestral statutes cannot be altered” and allied with the Empress Dowager, palace women, and eunuch forces, abolished all new policies immediately upon Shenzong’s death. The reformer was expelled from the court; the fruits of reform were reduced to zero.

- **Dynamical Diagnosis:** Wang Anshi used every legitimate tool the old system provided, and even created new institutions (the Finance Planning Commission) to bypass the old bureaucratic network. But he could not alter one fundamental fact: the personnel of the new institutions still came from the old system, and the implementation of the new policies still depended on the old network. When reform touched the fundamental interests of the parasites, immune rejection was triggered, and the sole weapon in Wang Anshi’s hand—imperial power—vanished with Shenzong’s death. **“When the man perishes, his policies perish” is the precise dynamical expression of the impossibility of institutionalizing the fruits of intra-systemic reform at the level of systemic rules.**

Case Two: Zhang Juzheng’s Reforms in the Mid-Ming Dynasty (1573–1582)

- **Reform Objectives:** Discipline the bureaucracy, survey landholdings, reform the tax system, consolidate border defenses.
- **Tools Employed:** The authority of the Senior Grand Secretary; the Performance Evaluation Method (*Kaocheng fa*) as an instrument of administrative oversight; the Single Whip Method (*Yitiao bianfa*) as an instrument of tax simplification; the trust of the Empress Dowager during the Wanli Emperor’s minority.
- **Parasite Network:** A vast, inert bureaucratic apparatus; powerful landlords colluding with local clerks; eunuch and consort clan forces at court.
- **Process of Dynamical Attenuation:** The Performance Evaluation Method, under Zhang Juzheng’s forceful imposition, temporarily raised administrative efficiency, but its essence was to constrain the bureaucracy through tighter documentary control rather than to alter the bureaucracy’s own interest structure. The Single Whip Method simplified taxation, but in implementation, local officials, having merged various surcharges into the regular tax, proceeded to impose new surcharges on top of the regular tax; the peasants’ burden did not fall but rose. **The reforms achieved localized technical successes but failed to touch the fundamental extraction mechanisms of the parasite network.**
- **Immune Rejection:** After Zhang Juzheng’s death, the anti-reform bureaucratic faction launched a comprehensive counteroffensive: Zhang Juzheng was posthumously stripped of his titles and honors, his property was confiscated, the Performance Evaluation Method was abolished, and the Single Whip Method gradually deviated in implementation. The Ming fiscal and administrative systems swiftly reverted to their old trajectories.
- **Dynamical Diagnosis:** Zhang Juzheng’s reforms were a classic instance of “person-dependent reform”—their entire efficacy was tethered to the reformer’s personal authority and the monarch’s forbearance. Once personal authority vanished, the system automatically reverted to its dominant direction of self-organization. **This is not a tragedy of Zhang Juzheng’s personal fate, but a dynamical proof that intra-systemic reform cannot solidify its achievements into new systemic rules.** The reformer may temporarily suppress the parasites, but cannot excise them—because the tools required for excision (institutionalized checks on power, independent oversight systems, bottom-up feedback channels) are precisely what the old system refuses to provide.

Case Three: The Gracchan Reforms in the Late Roman Republic (133–121 BCE)

- **Reform Objectives:** Limit the amount of public land (*ager publicus*) that could be held by any individual, redistribute excess land to landless citizens, and thereby restore the class of citizen-smallholders that formed the backbone of the Roman army.
- **Tools Employed:** The tribunician power to propose legislation; the voting procedures of the plebeian assembly; all legitimate means provided by the Roman republican constitution.
- **Parasite Network:** The senatorial aristocracy (*nobiles*). Through long-term occupation and de facto control, they had converted public land, legally owned by the state, into de facto private estates, and had deeply entrenched their control over the Senate, the Republic's central institution of power.
- **Process of Dynamical Attenuation:** Tiberius Gracchus, scrupulously following republican constitutional procedures, submitted his land bill to the plebeian assembly for a vote. The senatorial aristocracy induced another tribune, Octavius, to exercise his veto. Tiberius then had the assembly depose Octavius as a countermeasure—an act that itself touched the ambiguous boundary of the principle of “tribunician sacrosanctity” within the republican constitution, furnishing the aristocracy with the pretext that he was “destroying the Republic.” The bill passed, but the Senate, by controlling fiscal appropriations (denying operational funds to the land commission), paralyzed its implementation. When Tiberius sought re-election as tribune to continue his reform efforts, the senatorial aristocracy lynched him on the charge of “aspiring to kingship.” A decade later, his brother Gaius Gracchus, pushing through more extensive reforms, was ultimately suppressed by military force under the Senate's *senatus consultum ultimum* and took his own life.
- **Dynamical Diagnosis:** The Gracchan reforms represent the extreme of “legalism”—they employed every legitimate tool the republican constitution provided and strictly observed procedure. Yet when reform touched the fundamental interests of the parasites, the parasite network did not hesitate to **transgress the boundary of legality itself**, destroying the reformers through mob violence and military force. This case reveals a brutal dynamical fact: **the boundary within which legal tools may be used is precisely demarcated by the power of the parasites to define “legality.”** When the reformer attempts to use legal tools to alter the very framework of legality that the parasites define, the parasites will declare the reformer “illegal.” The tools cannot protect the tool-user.

Case Four: Gorbachev's Reforms in the Soviet Union (1985–1991)

- **Reform Objectives:** Revitalize the ossified Soviet system through *glasnost* (openness) and *perestroika* (restructuring), correct accumulated maladies, and achieve the self-renewal of socialism.
- **Tools Employed:** The authority of the CPSU General Secretary; the reformist consensus within the Party; the decision-making procedures provided by the constitution.
- **Parasite Network:** The bureaucratic privileged stratum (*nomenklatura*) that had formed over decades. This stratum enjoyed monopolies over information, resource allocation, and the attendant gray benefits within the planned economy, yet they did not truly own the means of production—their privileges depended upon their positions within the Party-state apparatus.
- **Process of Dynamical Attenuation:** Gorbachev sought to use *glasnost* to release societal

pressure and force the bureaucracy to accept reform. However, the criticism unleashed by *glasnost* rapidly exceeded the controllable scope of the reformers—it targeted not only specific policy errors but the historical legitimacy of the system itself. Simultaneously, amid the uncertainty of reform, the bureaucratic privileged stratum accelerated the conversion of their power into inheritable private wealth (“spontaneous privatization”). Parasitic behavior shifted from covert to overt, from slow extraction to frenzied plunder. **The tool of reform (openness of public discourse) did not weaken the parasites but accelerated their transformation—from covert parasites within the system to overt plunderers carving up state assets.**

- **Systemic Collapse (Rather Than Successful Reform):** When the reforms triggered a positive-feedback destabilization that the old system could not absorb, Gorbachev faced a choice: either use coercive force to terminate the reforms (returning to the old trajectory) or accept the disintegration of the system. He chose the latter. The Soviet Union was not repaired by reform; it collapsed from the systemic destabilization that reform triggered.
- **Dynamical Diagnosis:** Gorbachev’s reforms prove a yet deeper boundary: **in certain systems, the parasite network and the system’s meta-rules have become so deeply entangled that excising the parasites would necessarily entail the death of the system.** At this point, reform is no longer possible—because there is no “healthy tissue” left to preserve. Any move by the reformer is either rebuffed by the parasites (as in the Prague Spring of 1968) or triggers systemic collapse (as in the Soviet Union of 1991). The window for moderate reform, in such systems, has already closed.

The Common Structure Across Four Cases

Comparative Dimension	Wang Anshi’s Reforms	Zhang Juzheng’s Reforms	Gracchan Reforms	Gorbachev’s Reforms
Source of Reform Tools	Imperial edicts, bureaucratic apparatus	Grand Secretariat authority, Performance Evaluation Method	Tribunician legislation, plebeian assembly	General Secretary authority, intra-Party procedures
Controller of Tools	Emperor (nominal), bureaucratic network (actual)	Grand Secretariat (nominal), bureaucratic apparatus (actual)	Plebeian assembly (nominal), Senate (actual)	General Secretary (nominal), privileged stratum (actual)
Parasite Counter-	Implementation distortion,	Policy death with	Procedural obstruction,	Accelerated plunder,

Comparative Dimension	Wang Anshi's Reforms	Zhang Juzheng's Reforms	Gracchan Reforms	Gorbachev's Reforms
Strategy	discursive counter-attack, coup-style policy abolition	reformer's demise, procedural restoration	violent suppression	triggering systemic collapse
Outcome upon Touching Boundary	All new policies abolished	Reform reduced to zero	Reformers executed	System disintegrated
Attributable to Contingency?	No—isomorphic failure across eras	No—inevitable fate of person-dependent reform	No—legal tools cannot defeat monopolizers of legitimacy-definition	No—parasites deeply entangled with systemic meta-rules

Conclusion of Comparative Case Analysis: Four reform efforts, spanning three continents, two millennia, and three distinct institutional types (imperial, republican, single-party), exhibit a startling structural isomorphism in their modes of failure—**reform tools controlled or distorted by parasites, immune rejection triggered upon touching fundamental rules, failure to institutionalize reform achievements as new rules**. This cross-civilizational, cross-epochal isomorphism cannot be explained by “contingent errors” or “unripe conditions.” Dynamical inevitability is presently the only framework capable of furnishing a unified explanation.

5.4 Preemptive Response to a Potential Objection: Why Is Reform Initiated by the System Builder Also Infeasible?

The reader may at this point raise a natural objection: the reformers in the above cases—Wang Anshi, Zhang Juzheng, the Gracchi, Gorbachev—were not the supreme builders of their respective systems. They were participants acting within given rules (even if positioned at high levels). What, then, **if the system builder personally (e.g., an enlightened monarch) initiates top-down reform? Can this circumvent the boundary revealed by Corollary D17.1?**

This objection must be taken seriously. Intuitively, the system builder appears to possess an advantage that participants lack: they define the rules, and therefore ought to be able to modify them. History does indeed furnish cases of monarchs proactively driving reform. What must be demonstrated is this: **even when the system builder personally initiates reform, dynamically, its effects remain constrained by the same reform boundary**. The reasons are threefold.

First, the independence of the system's self-organization tendency.

Once a system is built and has operated to a steady state, it engenders a **self-organization tendency** (Axiom 3) that is independent of the builder's personal will. The builder may be the definer of the rules, but they are not the sole executor of the rules, still less the direct operator of every

subsystem within the system. The larger and more complex the system, the more severe the attenuation of the builder's personal will as it traverses layer upon layer of subsystems. In political systems, this self-organization tendency concretely manifests as the subsystem network that maintains the established pattern of interests—that is, the systemic parasites and their symbiotic structures. The builder may issue a reform edict, but they cannot personally deliver that edict to every village or supervise its execution by every official. They must rely upon the existing bureaucratic network. And the bureaucratic network—as previously demonstrated—is itself already a parasite or has been deeply captured by parasites.

Second, the transmission attenuation of reform directives.

The builder's reform directives, in traversing the subsystems, necessarily undergo **dynamical attenuation**. The mechanisms of attenuation operate on three levels: (1) **Information filtering**: Subsystems selectively feed back information to the builder regarding the implementation of reforms, systematically exaggerating successes and concealing distortions, so that the builder inhabits a cognitive illusion of “reform progressing smoothly.” (2) **Selective implementation**: Reform provisions favorable to the subsystem's own interests are actively implemented; provisions unfavorable are shelved, delayed, or creatively reinterpreted. (3) **Reverse utilization**: Subsystems convert the tools of reform into new instruments for enhancing their own extractive capacity. Wang Anshi's Green Sprouts Policy and Zhang Juzheng's Single Whip Method both suffered this reverse utilization at the local level. The builder may dismiss individual officials, but cannot simultaneously replace the entire bureaucratic network—and even if replaced, the newly appointed officials will, within the same systemic rules and interest structure, be rapidly assimilated.

Third, the sole means of bypassing the subsystems is to become a new builder.

If the builder seeks to bypass the already-parasitized subsystems and act directly upon the system's fundamental rules, the only feasible means is to **utterly destroy the old network and replace it with an entirely new organizational architecture**. But the dynamical essence of such an action is no longer “reform” but **systemic reconstruction**—the actor is no longer the builder of the old system but has become the **builder of a new system**. This is precisely the first path of transformation described earlier.

History furnishes clear contrasts. Shang Yang's reforms succeeded not because he skillfully reformed within the institutional framework of the old Qin state, but because he utterly destroyed the hereditary aristocracy's system of ranks and emoluments, replacing it with a wholly new rule—the system of military merit—that redefined Qin's social structure. The First Emperor of Qin's abolition of enfeoffment and establishment of commanderies and counties was likewise not a reform of the Zhou system but its systematic replacement. They were not reformers of the old system but builders of a new one.

Consider, by contrast, those builders who attempted self-renewal within the old systemic framework—such as Empress Dowager Cixi's tacit approval of the Self-Strengthening Movement or the Guangxu Emperor's pursuit of the Hundred Days' Reform in the late Qing. The efficacy of their reforms was strictly confined within the safe zone tolerated by the parasites. The Self-Strengthening Movement could introduce modern arsenals and steamships at the technical level, but the moment it touched the political system (the Hundred Days' Reform), the parasite network (the conservative faction centered on Cixi) immediately activated immune rejection and violently terminated the reform.

Conclusion: Reform initiated by the system builder is dynamically equivalent to intra-systemic reform. The tools the builder uses to implement reform are precisely the system's endogenous tools,

which have already been captured by parasites. If the builder insists on using these tools, the effects of reform will inevitably converge within the boundary demarcated by Corollary D17.1. If the builder attempts to destroy these tools, the action has already qualitatively transformed into systemic reconstruction and no longer falls within the category of “moderate reform.” The so-called “third path” of “reform initiated by the builder” is dynamically a pseudo-proposition.

5.5 The Theoretical Implication of the Reform Boundary: Distinguishing Reformists from Reconstructionists

Having established Corollary D17.1, an important conceptual distinction can now be clarified: **Reformists and reconstructionists belong to dynamically distinct categories.**

- **Reformists** attempt to correct the system’s maladies from within, using the legitimate tools the system provides. Their actions are constrained by Corollary D17.1; their effective domain is strictly confined to the safe zone tolerated by the systemic parasites.
- **Reconstructionists (system rebuilders)** attempt to become new system builders, replacing the old meta-rules with new ones. Their actions are not constrained by Corollary D17.1, but they are no longer acting within the category of “intra-systemic reform”—they are engaged in systemic reconstruction.

The practical implication of this distinction is clear: **If one believes that the existing system can achieve fundamental transformation through internal reform, one is dynamically mistaken, and one’s efforts will converge within the boundary tolerated by the system.** If one is committed to fundamental transformation, one must soberly recognize that the task is not to optimize the old system but to become the builder of a new system—which means one must be prepared to cultivate new cognitive frameworks, new organizational forms, and new legitimizing discourses at the system’s periphery, until the critical threshold of phase transition is reached.

This is not an ideological stance but a dynamical judgment.

Having demonstrated the dynamical boundary of reform, the next section turns to the future-oriented dimension of the new paradigm: **predicting the dual-threshold condition for civilizational phase transition and specifying the dynamical path of the leap from “collective mentality” to “metacognitive collective intelligence.”** If the preceding five sections have accomplished the critique of the old paradigm and the laying of the new paradigm’s foundation, then the sixth section shall accomplish the new paradigm’s ultimate mission—not to explain the cycle of the past, but to open the possibility of the future.

Part Six: Phase Transition Conditions and Future Prospects

The preceding five sections have accomplished a dual task: burying the old paradigm (by exposing its threefold blindness) and laying the foundation for the new (axiomatic foundation, core diagnostic criterion, the boundary of reform). Yet a dynamical theory that can only explain the past remains merely an elevated form of historiography. The ultimate mission of the new paradigm is to open a domain that the old paradigm never even approached—**prediction.**

If civilization is not doomed to cycle eternally, what are the conditions for breaking the cycle? If the individual is not a passive bearer of fate, how do individuals converge into a force capable of rewriting

the rules? This section shall answer both questions directly. We shall first deduce from theory the dual-threshold condition for civilizational phase transition, then offer a sober assessment of contemporary civilization's position on the phase transition spectrum, and finally specify a program of action that moves from “interpreting the world” to “changing it.”

6.1 The Theoretical Conditions for Phase Transition: From Individual Escape to Collective Emergence

In *Civilization Dynamics*, the phase transition from “Collective Mentality” to “Metacognitive Collective Intelligence” is described as a modelable dynamical process. The core criterion of this process is the **dual-threshold condition**:

Civilizational phase transition is triggered if and only if the proportion of second-order noospheric humans (awakened individuals) in the total population, X , satisfies $X \geq X_c$, and the degree of societal narrative receptivity, Ξ , simultaneously satisfies $\Xi \geq \Xi_c$.

The following subsections explicate the dynamical meaning of each variable and its critical value in turn.

The First Threshold: The Proportion of Awakened Individuals, X

Definition. X is the proportion of second-order noospheric humans within the system. A second-order noospheric human (see Core Terminology for full definition) does not refer broadly to “educated persons” or “elites,” but specifically to individuals who simultaneously satisfy the following three conditions:

1. **Metacognitive capacity M has reached a critical level:** The individual can consciously observe, compare, and switch between the cognitive frameworks they employ, rather than being unconsciously dominated by a single framework.
2. **Cognitive escape from the dominant narrative has been completed:** The individual no longer regards the system's dominant narrative as the sole “reality” and can recognize that narrative's constructed and constraining character.
3. **Active integration into a synergetic cognitive network:** The individual consciously establishes informational coupling with other awakened individuals at the system's periphery and participates in the collective iteration of cognitive protocols.

Why second-order noospheric humans rather than other groups? The old paradigm's subjects of transformation were typically “oppressed classes” or “elite reformers.” The new paradigm's subject of transformation is the **cognitive escapee**. The rebellion of the oppressed may overthrow the old system, but if their cognitive framework remains trapped within the old system, the new system will merely replicate the structure of the old—this is the dynamical root of why peasant rebellions in the dynastic cycle, even when successful, established dynasties of the same type. Elite reformers, as demonstrated in the previous section, are constrained by the boundary of intra-systemic tools. Only second-order noospheric humans, because their cognitive framework has already transcended the semantic field of the old system, are capable of designing genuinely heterogeneous new rules.

Estimation of the critical value X_c . Based on the universal regularities of phase transitions in complex systems (percolation models, opinion dynamics models) and limited induction from historical cases, *Civilization Dynamics* proposes $X_c \approx 21\%$ (approximately one-fifth). The theoretical basis for this figure is as follows: in opinion dynamics, when the proportion of committed minority holders exceeds approximately 20%–25%, and provided the social network satisfies basic connectivity, the

minority viewpoint undergoes a phase transition from “invisible peripheral noise” to “an undeniable candidate for social consensus.” This is not a precise constant but a dynamical range. The core contribution of this paper lies in establishing the criterion that **a critical threshold exists**; its precise numerical value awaits future empirical calibration.

Crucial Clarification: X must not be conflated with rates of higher education. Theorem B3 (Specialization as Cognitive Differentiation) has already demonstrated that specialized education, while enhancing an individual's cognitive acuity within a particular domain, necessarily degrades their capacity to perceive other parts of the system. A highly specialized Ph.D. may, outside their narrow domain of expertise, remain a **standard replicant**—they excel at optimization within a particular sub-framework but have never examined the framework itself. The expansion of higher education, in the absence of metacognitive training, may **decrease** rather than increase X , because it molds more individuals into highly efficient differentiated cells, embedding them ever deeper within the gravitational well of a single cognitive framework. The measurement of X must be based on direct assessment of metacognitive capacity (e.g., lens-switching tests, framework audit interviews), not on educational credentials or knowledge tests.

The Second Threshold: Societal Narrative Receptivity, Ξ

Definition. Ξ is the collective perception within a society of the degree to which the old narrative has failed, and the intensity of demand for a new narrative. Its value ranges from 0 to 1. The higher Ξ is, the more willing society is to accept a reinterpretation of reality; as Ξ approaches zero, even if awakened individuals propose a new framework, society as a whole will dismiss them as “lunatics” or “heretics” and ignore them.

The constituent factors of Ξ . Ξ is not a mysterious *Zeitgeist* but the product of three observable variables:

- **The systemic parasite burden rate, Y :** When the proportion of social resources appropriated by parasites exceeds the threshold of individual tolerance, a vast chasm opens between the old narrative's promise (“hard work leads to a good life”) and the individual's lived experience (“no matter how hard I work, I am plundered”). The higher Y is, the deeper the chasm, and the larger Ξ becomes.
- **The intensity of cognitive dissonance, C :** When the old framework repeatedly fails to explain reality, cognitive dissonance arises within the individual. C is the intensity of dissonance at the individual level; its societal aggregate constitutes the micro-foundation of Ξ (Theorem D23).
- **The accessibility of alternative narratives:** Even if the old narrative has failed, if no comprehensible alternative narrative is accessible, society may fall into a state of disorientation—“knowing the old is wrong but not knowing where the new lies”—and Ξ may still fail to reach the critical threshold. The accessibility of alternative narratives depends on the cognitive fluidity constant $*k*$ (the degree to which civilizational infrastructure facilitates cognitive flow).

The critical value Ξ_c . The precise determination of Ξ_c is more complex than that of X_c as it involves the aggregation of subjective perception. However, theory can provide a qualitative criterion: when mainstream public discourse shifts from “individual problems can be fixed” to “the system itself must be rewritten,” Ξ is approaching the critical threshold. Historical markers include: large-scale social movements that no longer make specific policy demands but instead question fundamental institutions; core symbols of the old narrative (e.g., “ancestral statutes,” “free markets”) shifting from

sacred and unquestionable to objects of open debate; the old system's suppression of heterodox narratives shifting from “indifference” to “panicked repression.”

The Logical Relation Between the Two Thresholds: Why Must Both Be Satisfied Simultaneously?

$X \geq X_c$ and $\Xi \geq \Xi_c$ are both necessary; neither is sufficient alone. Their dynamical division of labor is as follows:

- **X provides the “cognitive supply”:** Awakened individuals are the creators and carriers of new cognitive frameworks. Without sufficient X , new frameworks cannot attain stable expression within the system.
- **Ξ provides the “cognitive demand”:** Societal suffering is the collective signal that the old framework has failed. Without sufficient Ξ , new frameworks, even if proposed, will be rejected by the societal immune system.

History repeatedly corroborates this division of labor. In the late Ming period, the intellectual breakthroughs of Huang Zongxi, Gu Yanwu, and Wang Fuzhi (local emergences of X) had already reached fundamental critiques of monarchy, but the overall societal Ξ had not yet reached the critical threshold—the inertia of the old narrative and the exogenous shock of the Manchu conquest together suppressed the rise of Ξ . These ideas were buried for over two centuries, only to be reactivated as noospheric gene reserves for transformation when Ξ surged to the critical threshold in the late Qing. Conversely, on the eve of the French Revolution, Ξ was extraordinarily high due to the parasite burden of the Old Regime (fiscal bankruptcy, rampant privilege), but without the decades-long accumulation of X produced by the Enlightenment (awakened individuals such as Voltaire, Rousseau, and Diderot, and the synergetic cognitive networks they constructed), the energy of Ξ would have been released merely as riot and would not have crystallized into the new system's meta-rules.

The theoretical significance of the dual-threshold condition is this: it furnishes, for the first time, an analyzable dynamical framework for why revolutions succeed at certain moments and fail at others. Revolution is not a random occurrence, nor is it determined solely by “oppression”—if oppression determined revolution, human history would be replete with revolutions. Revolution is a phase transition event at the intersection of two curves, X and Ξ , on the axis of time.

6.2 The Position of Contemporary Civilization on the Phase Transition Spectrum: A Sober Assessment

Having established the theoretical conditions for phase transition, a natural question arises: **How close is our own era to the critical threshold of phase transition?**

The answer provided in this section will maintain methodological sobriety. The core contribution of this paper is the theoretical framework of the phase transition condition itself, not the claim to have precisely measured current global values of X and Ξ . The following assessment is a qualitative interpretation of observable trends through the lens of the theoretical framework; its value lies in **indicating the direction for measurement** rather than in delivering definitive conclusions.

Trend One: Factors Potentially Increasing X (Proportion of Second-Order Noospheric Humans)

The following factors may be increasing global X :

1. **The surge in the cognitive fluidity constant k due to information infrastructure.** The internet, especially decentralized social media and open-source knowledge platforms, has reduced the cost of disseminating heterodox cognitive frameworks to a historic low. A

peripheral cognitive probe (such as an independent researcher or a small heterodox intellectual community) can now reach a global audience without passing through traditional gatekeepers of authority (academic journals, official media). This dramatically raises k , thereby lowering the threshold for individual cognitive escape (Theorem D23: $P(\text{escape}) = k \cdot M \cdot C / (A \cdot N)$).

2. **The proliferation of cross-framework experiences.** Globalization, migration, transnational education, and interdisciplinary work have forced or enticed increasing numbers of individuals into exposure to multiple, heterogeneous cognitive frameworks. Such exposure is a natural training ground for metacognitive capacity M —when a person personally experiences that “the same event is interpreted in entirely different ways in different cultures or disciplines,” their belief in the absoluteness of their original framework undergoes a loosening (Theorem B10: The Meta-Framework Loosening Effect of Philosophical Reflection).
3. **The manifest failure of old frameworks.** Climate crisis, global pandemics, financial instability, and persistent inequality—these global challenges resist convincing solution within the old frameworks (whether neoliberal market fundamentalism or traditional developmental statism). The exhaustion of the explanatory power of old frameworks prompts more individuals to shift from “optimization within the framework” to “searching outside the framework.”

Trend Two: Factors Offsetting or Suppressing the Growth of X

The following factors may be suppressing or offsetting the growth of X :

1. **The cognitive differentiation effect of specialized education** (Theorem B3). The massification of global higher education has turned more individuals into highly efficient differentiated cells. A machine learning engineer in Silicon Valley, a quantitative analyst on Wall Street, a mergers and acquisitions lawyer at a transnational firm—their cognition is highly sharpened within their professional domain, yet they may lack basic metacognitive awareness of the operating logic of the system as a whole. They are not awakened individuals but the most efficient cogs of the old system. Years of education, in the absence of metacognitive training, may be negatively correlated with X .
2. **The cognitive cocoon effect of algorithmic recommendation.** While information infrastructure has raised k^* , commercial algorithmic recommendation systems, through positive feedback mechanisms, continuously expose individuals to information streams highly congruent with their existing frameworks. This artificially inflates the paradigm's authority A and reduces the natural triggering frequency of cognitive dissonance C . Individuals are not forcibly indoctrinated with a single framework but are “gently” locked into comfort zones by algorithms, losing the opportunity to encounter heterogeneous frameworks and generate cognitive dissonance.
3. **The large-scale deployment of cognitive weapons** (Theorem F37). State and non-state actors are becoming increasingly adept at using cognitive weapons—through targeted narrative deployment, symbol manipulation, and attention hijacking—to guide or suppress the expression of particular cognitive frameworks. Cognitive deadlock munitions (which directionally increase A and N) are widely used to maintain the stability of established paradigms. In certain societies, this artificially suppresses the growth of X and channels the rise of $\mathcal{E}$ into harmless venting rather than fundamental reflection.

Synthetic Judgment: Global X is currently caught in a tug-of-war between opposing forces. The

openness of information infrastructure and the proliferation of cross-framework experiences provide historically unprecedented opportunities for cognitive escape; but the deepening of specialized education, the lock-in of algorithmic cocoons, and the deployment of cognitive weapons constitute unprecedented obstacles to escape. **There is no basis for assuming that X will automatically grow to the critical threshold.** Phase transition is not a historical inevitability but the outcome of a dynamical contest.

Trend Three: Indicators of Rising $\mathcal{E}$ (Societal Narrative Receptivity)

In contrast to the contested state of X , global $\mathcal{E}$ exhibits a clearer upward trend. The following phenomena may be regarded as proxy indicators of rising $\mathcal{E}$:

1. **The collapse of trust in old narratives.** Globally, public trust in traditional authoritative institutions—governments, media, large corporations, international organizations—is in sustained decline. This is not an isolated phenomenon in a few countries but a cross-civilizational trend. When the core carriers of the old narrative (authoritative institutions) lose trust, the old narrative itself loses the foundation of its credibility.
2. **The explicit emergence of “system rewrite” demands.** From Occupy Wall Street to the *Gilets Jaunes* movement, from global climate strikes to various forms of social protest, an increasing number of collective actions no longer merely make specific policy demands (“cut taxes,” “raise wages”) but instead question the rules of the system itself (“the game is rigged,” “this system is unsustainable”). This is a qualitative marker that $\mathcal{E}$ is shifting from “individual correction” to “system rewrite.”
3. **The market testing of alternative narratives.** Across the world, political forces, economic models, and modes of life heterodox to the mainstream narrative are undergoing large-scale social experimentation. Regardless of the specific outcomes of these experiments, their very existence proves that society as a whole is no longer satisfied with the single option provided by the old narrative and is actively seeking or testing alternatives. This is a strong signal that $\mathcal{E}$ has entered a high range.

Synthetic Judgment: $\mathcal{E}$ may already be approaching or within the critical range in multiple societies. The explanatory and promissory power of old narratives has been severely overdrawn, and societal demand for “rewriting the rules” has never been clearer. However, high $\mathcal{E}$ without sufficient X to match it will release its energy in destructive rather than constructive forms—this is precisely why most collapses throughout history have not led to phase transition but merely to another round of the cycle.

A Dynamical Diagnosis of the Present Moment

Based on the foregoing assessment, the following sober judgments may be offered regarding the position of contemporary civilization on the phase transition spectrum:

1. **$\mathcal{E}$ has likely approached or entered the critical range.** The systemic failure of old frameworks is no longer a peripheral perception but a widespread societal recognition.
2. **X is in a state of contestation; there is not yet evidence that it has breached the critical threshold.** The escape opportunities of the information age coexist with new forms of lock-in; the net direction of X is uncertain.
3. **The moment of dual-threshold convergence has not yet arrived, but it may be approaching.** If the growth factors of X prevail over the offsetting factors in the long-term contest, the mid-21st century could become a critical window for phase transition. If the

offsetting factors dominate, civilization may enter a protracted period of “high Ξ , low X ” turbulence—society knows the old is broken but does not know where the new lies, and thus continues to expend itself in cycles of collapse and reconstruction.

Reiteration of Methodological Note: The above judgments are not empirical conclusions but qualitative interpretations of observable trends through the lens of the theoretical framework. Their value lies in furnishing a set of testable hypotheses for future empirical research—for example, can measurement instruments for X and Ξ be developed? Can the dual-threshold condition be retroactively verified against historical data? These are precisely the research tasks of the fourth sub-paper (Future Prospects).

6.3 From Theory to Action: The Practical Roadmap Provided by the New Paradigm

If the analysis of this paper is correct—the dynastic cycle is the dynamical product of cognitive lock-in and parasitic proliferation, moderate reform confronts an impassable boundary, and phase transition requires the convergence of the dual thresholds of X and Ξ —then what should an individual situated within the system **do**?

The old paradigm has no answer to this question. It can only offer two traditional scripts: either strive to rise within the system (become a more efficient cog), or rise in rebellion when the system collapses (become fuel for the next round of the cycle). The new paradigm, because it reveals the dynamics of individual cognitive escape and the emergence of synergetic networks, provides a third path.

First Principle: Enhance metacognitive capacity M ; complete individual cognitive escape.

This is the absolute prerequisite for all subsequent action. An individual who has not yet completed cognitive escape, no matter how morally upright or resolute in will, remains within the gravitational field of the old system in their cognitive framework. Their “resistance” will unconsciously employ the symbols and logic provided by the old system, and thus will replicate the old system at a deep structural level.

Concrete paths for enhancing M include but are not limited to:

- **Lens matrix training** (Theorem B6): Consciously employ multiple analytical lenses (systemic lens, power lens, evolutionary lens, narrative lens, etc.) to deconstruct the same phenomenon, practicing rapid switching between frameworks.
- **Actively generating cognitive dissonance C** : Continuously expose oneself to high-quality heterogeneous information that conflicts with one's existing frameworks, rejecting the cognitive comfort zone constructed by algorithmic recommendation.
- **Philosophical reflection exercises** (Theorem B10): Subject taken-for-granted concepts to “defamiliarizing” questioning, loosening the rigidity of the meta-framework.

Individual cognitive escape is not a one-time epiphany but continuous self-training. Its goal is not to find an “ultimate true framework” but to acquire **the capacity to switch freely among multiple frameworks without being captured by any single one.**

Second Principle: Join or create synergetic cognitive networks.

Individual cognitive escape is lonely. But isolated escapees cannot change the system—they can only become marginalized “hermits” or “heretics.” Changing the system requires **synergetic cognitive networks**: information-coupling structures formed among multiple second-order noospheric humans, within which new cognitive frameworks can be verified, iterated, and strengthened.

In the contemporary era, synergetic cognitive networks can take many forms: open-source knowledge communities, interdisciplinary research groups, distributed autonomous organizations, intellectual co-

creation circles. Their common features are:

- Members share the metacognitive foundation of “awareness of frameworks”;
- The network internally operates according to protocol modes rather than command modes (Theorem A0);
- The output of the network is not a unified dogma but continuously evolving cognitive protocols.

Joining a synergetic cognitive network transforms individual cognitive escape from a private experience into collective intelligence. Creating a synergetic cognitive network is the most direct contribution a second-order noospheric human can make to civilizational phase transition.

Third Principle: Construct prototypes of new systems at the periphery of the old system.

Corollary D17.1 has rigorously proven that intra-systemic reform cannot reach the rewriting of fundamental rules. But this does not mean that awakened individuals can only wait for the old system to collapse. The evolutionary path of the new paradigm (the second path of phase transition) requires awakened individuals to construct **operational prototypes** of new systemic rules at the **periphery** of the old system—in domains where the old system's control is weak.

These prototypes need not attempt to replace the old system immediately. Their functions are:

- **To demonstrate the technical feasibility of new rules:** to show that another way of organizing production, distribution, and decision-making is possible.
- **To accumulate a noospheric gene reserve of new rules:** to encode new rules into replicable and transmissible symbols, protocols, and tools.
- **To provide an accessible alternative when Ξ in the old system continues to rise:** so that when the old narrative collapses, society does not fall into the disorientation of “not knowing where the new lies.”

Historically, the “Invisible College” of the Scientific Revolution, the salons and correspondence networks of the Enlightenment, and the distributed collaboration communities of the open-source software movement all played, to varying degrees, this role of “peripheral prototypes.” They did not directly overthrow the old system, but they cultivated the seeds of the new system at the old system's periphery. When Ξ breached the critical threshold, these seeds acquired the social soil for large-scale growth.

Fourth Principle: Reject the illusion of reform; invest finite energy into effective action.

This is the practical corollary of Corollary D17.1. If moderate reform is dynamically doomed to fail, then continuing to invest personal energy into intra-systemic reform—lobbying within existing political procedures, debating in mainstream media, seeking “corrections” within the framework of old institutions—is not prudence but complicity with the pathology. Each failed reform not only expends the energy of awakened individuals but, more importantly, depletes society's **hope for change**—it teaches more people that “change is impossible,” thereby suppressing the rise of Ξ and the convergence of X .

Awakened individuals should withdraw their finite life-energy from doomed reform attempts and redirect it toward the actions indicated by the preceding three principles: enhancing M , joining networks, constructing prototypes. This is not passive withdrawal but active strategic redeployment—from the core zones where the old system's gravitational field is strongest to the peripheral zones where new systems may emerge.

6.4 Preemptive Response to the Charge of “Elite Betrayal”

After setting forth the above program of action, a possible criticism is this: Does this not amount to encouraging awakened individuals to “flee” the system, abandoning the old system to the parasites and standard replicants, thereby constituting a betrayal by the elites?

This criticism merits a serious response.

First, cognitive escape is not equivalent to physical flight. An individual who has completed cognitive escape may perfectly well continue to reside physically within the old system—pursuing the same occupation, living in the same community. What has changed is the way they understand these activities: they no longer bind their life’s meaning to the evaluative criteria provided by the old system (promotion, titles, recognition), but re-anchor their actions in the value-coordinates of a new framework. They may work within the old system, but their energy is no longer directed at maintaining the old system but at nourishing the new prototypes at the periphery.

Second, the “exit” of awakened individuals is a symptom of systemic pathology, not its cause. If a system makes its most metacognitively capable members feel that effective change within the system is impossible, prompting them to construct new prototypes at the periphery, then the problem lies in the system’s cognitive lock-in, not in the choices of the awakened. To attribute the marginalization of the awakened to their “betrayal” is a classic instance of the old paradigm’s attributional inversion—once again shifting the blame for systemic pathology onto the individual.

Third, history proves that peripheral prototypes are the necessary incubators of civilizational phase transition. No one designed modern democracy through legal procedures in the Roman Senate. No one created the modern scientific method through scholastic disputation in medieval universities. The meta-rules of new systems have never been generated through reform at the center of old systems; they have emerged through exploration and trial-and-error at the periphery. The movement of awakened individuals toward the periphery is not a betrayal of civilization but an assumption of the function most needed by civilization on the eve of phase transition—**drawing the blueprints for a new world.**

6.5 From Foresight to Creation

This section has completed a full arc from theory to action. We began from the dual-threshold condition for phase transition, offered a sober assessment of contemporary civilization’s position, and finally provided a program of action for awakened individuals.

The old paradigm’s terminus is “interpreting the world”—it is content to explain why dynasties cycle. The new paradigm’s point of departure is “changing the world”—it shows how individuals can converge into a force that rewrites the rules.

In the next and final section, we shall review the argumentative arc of the entire paper and deliver the ultimate declaration: the dynastic cycle is not an iron law of history but a cage for the mind. The moment to shatter this cage lies not in the distant future, but in the instant each individual decides to see anew their relationship to the system.

Part Seven: Conclusion—From Fated Cycle to Conscious Design

The argumentative arc of this paper has now closed. In this concluding section, I shall accomplish three tasks: review the core argument of the paper as a whole, clarify the fundamental respects in which the new paradigm transcends the old, and finally return the theory to its ultimate owner—each individual

who reads these words.

7.1 Review of the Argument

This paper set out from a foundational question: If the “dynastic cycle” is a law that reveals historical regularity, what exactly does it explain?

Part One exposed the threefold blindness of the old paradigm. Its attribution stops at surface phenomena—blaming decline on institutional decay, decadent rulers, and foreign invasion—yet it cannot answer why institutions *must* decay, why wise rulers *must* be followed by foolish ones, and why foreign enemies *always* seem to exploit the moment of greatest systemic vulnerability. Its positioning of the individual erases agency—reducing billions of living, cognitive beings to passive input-output units of systemic functions, wholly ignoring the countervailing force that individual cognitive differentiation, escape, and convergence exert upon the direction of systemic evolution. Its path to change harbors an illusion of reform that has never been subjected to dynamical scrutiny—tacitly assuming that system participants can repair the system using the system's own tools, without ever asking whether those tools have themselves already been captured by parasites.

Part Two laid the axiomatic foundation for the new paradigm. Five axioms—Logical Self-Reference, Material/Noospheric Duality, Self-Organization Tendency, Relativity of Cognitive Frameworks, and Symbol as Noospheric Gene—constitute a logically coherent dynamical framework. Together they reveal that civilization is not merely an accumulation of matter and institutions but an ecology of cognitive protocols and symbolic systems. The pathology of civilization lies not at the material level but at the noospheric level—in the inevitable proliferation of systemic parasites, in the path-dependent lock-in of cognitive frameworks, and in the systematic obstruction of pathways for individual cognitive escape.

Part Three proposed a new diagnostic criterion for civilizational health: **Civilization Potential = $\Sigma(\text{Individual Niche Actualization Degree}) \times \text{Systemic Resource Fluidity} \times (1 - \text{Systemic Parasite Burden})$** . This criterion comprehensively surpasses the old metrics of GDP, territory, and longevity in explanatory power. It explains how “powerful” dynasties can collapse in an instant—because the latter two factors had long since approached zero, and the system had been running near the critical threshold all along. It places the fate of the individual at the center of civilizational evaluation for the first time—the strength of a civilization lies not in how many individuals it contains but in how many individuals it enables to live out their innate gifts.

Part Four rigorously demonstrated the dynamical boundary of moderate reform. Building upon Theorem D17 (System Builder's Advantage), this paper formally proposed **Corollary D17.1: The Boundary Theorem of Intra-Systemic Reform**. The theorem proves that in any system where the dominant direction of self-organization is defined by the system builder, any reform attempt reliant upon the system's endogenous tools will inevitably have its effects converge within the safe zone tolerated by the systemic parasites, and can never reach the rewriting of fundamental rules.

Comparative case analysis of reform efforts across Chinese and Western history—from Wang Anshi to Zhang Juzheng, from the Gracchi to Gorbachev—exhibited a startling structural isomorphism, providing empirical support for this dynamical boundary that spans civilizations and epochs.

Part Five predicted the dual-threshold condition for civilizational phase transition. The phase transition from “Collective Mentality” to “Metacognitive Collective Intelligence” depends upon the proportion of awakened individuals X and the degree of societal narrative receptivity Ξ simultaneously exceeding their critical thresholds. Based on complex systems theory, *Civilization Dynamics* proposes $X_c \approx 21\%$.

A dynamical diagnosis of contemporary civilization suggests that $\mathcal{E}$ may already be approaching the critical range, but X remains contested between opposing forces, and the moment of dual-threshold convergence has not yet arrived. The program of action for awakened individuals is thus clear: enhance metacognitive capacity M , complete individual cognitive escape; join or create synergetic cognitive networks; construct operational prototypes of new systems at the periphery of the old; withdraw finite energy from doomed reform attempts and redirect it toward these effective actions.

7.2 From Cycle to Phase Transition: The Fundamental Transcendence of the New Paradigm

The difference between the dynastic cycle and Civilization Phase Transition Dynamics is not merely a matter of explanatory power; it is a rupture between two fundamentally distinct philosophies of history.

The old paradigm is retrospective; the new paradigm is prospective. The old paradigm can only explain after the fact why a civilization collapsed—and it always “succeeds” at this, because its explanatory variables (institutional decay, ruler incompetence) are symptoms that any collapse inevitably presents. This is an unfalsifiable circular argument. The new paradigm furnishes criteria that can be measured *before* collapse—parasite burden rate, resource fluidity, individual niche actualization degree—and predicts, based on these criteria, when the system is approaching criticality. It accepts empirical testing and may therefore be falsified. This is the leap from historiography to dynamics.

The old paradigm is fatalistic; the new paradigm is action-oriented. The old paradigm tells people: rise and fall are inevitable; the individual is powerless. It presents history as a script already written, with humans merely the characters in the play. The new paradigm reveals: phase transition has conditions, and conditions can be intervened upon. It restores history from “script” to “battlefield”—a dynamical contest constituted by the convergence of countless individual cognitive choices. It tells people: the future is not fated; it is determined by the proportion of awakened individuals, the density of synergetic networks, and the threshold of parasite burden. And each of these variables can be altered by individual action.

The old paradigm erases the individual; the new paradigm empowers the individual. This is the most fundamental divergence between the two. The narrative of the dynastic cycle is intellectually impoverished and practically pernicious. It has taught generation after generation to believe that collapse is inevitable, that effort is ultimately futile. This belief itself serves as a catalyst for civilizational ossification—when the majority of a civilization's members cease to believe in their own capacity to rewrite the rules, that civilization has already entered cognitive heat death. The entire effort of the new paradigm is to dismantle this prison of thought. It proves that individual cognitive escape is a real dynamical possibility; that the convergence of awakened individuals can form a collective force capable of rewriting the rules; that the path by which civilization breaks the cycle has now been theoretically opened.

7.3 The Return of Theory: From Author to Reader

At the opening of this paper, I declared the reflexivity of this theory—it acknowledges that it too is subject to the relativity of cognitive frameworks; it is not a closed dogma claiming possession of absolute truth, but an open cognitive operating system capable of observing, revising, and iterating upon itself.

At the close of this paper, I must complete the final link of this reflexivity commitment: to return the theory to its ultimate owner.

The ultimate owner of theory is not the author. The author is merely the midwife of theory. A theory—

if it is alive—should begin its second life within each cognitive individual it reaches. The reader is not a consumer of theory but a co-evolutionary participant in theory. Every doubt you raise about this paper, every revision you propose, every attempt you make to test it in practice—all these are part of the life of this theory.

Therefore, the final conclusion of this paper is not a statement but an invitation.

If, after reading this paper, you begin to consciously observe the cognitive frameworks you employ—you have already set foot on the path of individual cognitive escape.

If you begin to seek out those who see the world differently from you yet provoke thought rather than mere irritation—you are already at the edge of joining a synergetic cognitive network.

If you begin, within your own domain, to experiment with new rules on a small scale—even if only the collaboration method of a work team, the decision-making process of a community, or an open-source protocol for knowledge production—you are already planting a seed for a new system at the periphery of the old.

These actions, within the narrative of the old paradigm, are utterly meaningless. They will not bring you more power, wealth, or recognition within the old system. But within the dynamics of the new paradigm, each of them contributes an infinitesimal increment to the growth of X , adds a faint signal to the rise of Ξ , and pushes civilization an imperceptible distance from this side of the critical threshold toward the other.

7.4 The Final Declaration

The era in which the dynastic cycle ruled human self-understanding should be declared closed at this moment.

It is not a law of history but a cage for the mind. The reason it appeared irresistible is not that it revealed truth, but that it succeeded in making enough people *believe* it—and thus enough people abandoned the attempt to rewrite the rules, and thus the systemic parasites could repeat their feast in every cycle.

To shatter this cage requires no waiting for a wise ruler, no expectation of a benevolent system, no investment of hope in any intra-systemic authority.

It requires only that each individual, at some moment, make a single decision: **to regard oneself no longer as a cog in the system but as an examiner of the rules; to entrust one's cognition no longer to any narrative that brooks no doubt but to make one's own metacognition the final arbiter; to wait no longer for others to change the world but to become, within one's own niche, a prototype of new rules.**

The equation of Civilization Phase Transition Dynamics has been given. The variable X —the proportion of awakened individuals—is distributed among every individual who reads these words.

You are not the bearer of the cycle.

You are the critical threshold of phase transition.

The moment of decision lies not in the future.

It lies in this very instant—the instant after you close this paper, when you decide how to understand your own relationship to the system.

References

I. Author's Core Theoretical Works

He, G. (2025). *Noospheric Ecology: A Systems Theory of Cognition, Culture, and Society*. Zenodo.

DOI: [10.5281/zenodo.19353340](https://doi.org/10.5281/zenodo.19353340)

He, G. (2025). *Civilization Dynamics: Planetary Civilizational Science*. Zenodo.

DOI: [10.5281/zenodo.19353434](https://doi.org/10.5281/zenodo.19353434)

II. Civilizational Morphology and Philosophy of History (Representative Works of the Old Paradigm)

Toynbee, A. J. (1934–1961). *A Study of History* (Vols. 1–12). Oxford University Press.

Spengler, O. (1918–1922). *Der Untergang des Abendlandes* [The Decline of the West]. C. H. Beck.

Gibbon, E. (1776–1789). *The History of the Decline and Fall of the Roman Empire*. Strahan & Cadell.

Ibn Khaldūn. (1377). *Kitāb al-ʿIbar* [The Book of Lessons]. (F. Rosenthal, Trans., 1958, *The Muqaddimah: An Introduction to History*. Princeton University Press).

Huang, Y. (1945). *Yan'an guilai* [Return from Yan'an]. Guoxun Shudian.

Sima, G. (1084). *Zizhi tongjian* [Comprehensive Mirror in Aid of Governance]. (Various historical editions).

III. Complex Systems and Phase Transition Theory

Bak, P. (1996). *How Nature Works: The Science of Self-Organized Criticality*. Copernicus.

Prigogine, I., & Stengers, I. (1984). *Order out of Chaos: Man's New Dialogue with Nature*. Bantam Books.

Haken, H. (1983). *Synergetics: An Introduction* (3rd ed.). Springer.

Kauffman, S. A. (1993). *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press.

Barabási, A.-L. (2002). *Linked: The New Science of Networks*. Perseus Publishing.

Watts, D. J. (1999). *Small Worlds: The Dynamics of Networks between Order and Randomness*. Princeton University Press.

Granovetter, M. S. (1973). The Strength of Weak Ties. *American Journal of Sociology*, *78*(6), 1360–1380.

Centola, D., & Macy, M. (2007). Complex Contagions and the Weakness of Long Ties. *American Journal of Sociology*, *113*(3), 702–734.

Xie, J., Sreenivasan, S., Korniss, G., Zhang, W., Lim, C., & Szymanski, B. K. (2011). Social Consensus through the Influence of Committed Minorities. *Physical Review E*, *84*(1), 011130.

IV. Cognitive Science and Metacognition Theory

Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.

Fleck, L. (1935). *Entstehung und Entwicklung einer wissenschaftlichen Tatsache* [Genesis and Development of a Scientific Fact]. Benno Schwabe.

Flavell, J. H. (1976). Metacognitive Aspects of Problem Solving. In L. B. Resnick (Ed.), *The Nature of Intelligence* (pp. 231–235). Lawrence Erlbaum.

Flavell, J. H. (1979). Metacognition and Cognitive Monitoring: A New Area of Cognitive-

- Developmental Inquiry. *American Psychologist*, *34*(10), 906–911.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Hofstadter, D. R. (1979). *Gödel, Escher, Bach: An Eternal Golden Braid*. Basic Books.
- Dennett, D. C. (1991). *Consciousness Explained*. Little, Brown and Company.
- Clark, A. (1997). *Being There: Putting Brain, Body, and World Together Again*. MIT Press.
- Dawkins, R. (1976). *The Selfish Gene*. Oxford University Press. (Especially Chapter 11, "Memes: The New Replicators")
- Blackmore, S. (1999). *The Meme Machine*. Oxford University Press.
- Dennett, D. C. (2017). *From Bacteria to Bach and Back: The Evolution of Minds*. W. W. Norton & Company.

V. Systems Theory and Social Systems Pathology

- von Bertalanffy, L. (1968). *General System Theory: Foundations, Development, Applications*. George Braziller.
- Luhmann, N. (1984). *Soziale Systeme: Grundriß einer allgemeinen Theorie* [Social Systems]. Suhrkamp.
- Olson, M. (1982). *The Rise and Decline of Nations: Economic Growth, Stagflation, and Social Rigidities*. Yale University Press.
- Tainter, J. A. (1988). *The Collapse of Complex Societies*. Cambridge University Press.
- Diamond, J. (2005). *Collapse: How Societies Choose to Fail or Succeed*. Viking Press.
- Scott, J. C. (1998). *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*. Yale University Press.
- Scott, J. C. (1985). *Weapons of the Weak: Everyday Forms of Peasant Resistance*. Yale University Press.
- Acemoglu, D., & Robinson, J. A. (2012). *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. Crown Business.
- Acemoglu, D., & Robinson, J. A. (2019). *The Narrow Corridor: States, Societies, and the Fate of Liberty*. Penguin Press.
- North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge University Press.
- North, D. C., Wallis, J. J., & Weingast, B. R. (2009). *Violence and Social Orders: A Conceptual Framework for Interpreting Recorded Human History*. Cambridge University Press.

VI. Chinese and Western Historical Case Studies

Chinese History

- Yu, Y. (2003). *Zhu Xi de lishi shijie: Songdai shidafu zhengzhi wenhua de yanjiu* [The Historical World of Zhu Xi: A Study of the Political Culture of Song Dynasty Literati]. Sanlian Shudian.
- Huang, R. (1981). *1587, A Year of No Significance: The Ming Dynasty in Decline*. Yale University Press.
- Deng, G. (1997). *Beisong zhengzhi gaigejia Wang Anshi* [Wang Anshi: Political Reformer of the Northern Song]. Hebei Jiaoyu Chubanshe.
- Zhu, D. (1945). *Zhang Juzheng dazhuan* [A Biography of Zhang Juzheng]. Kaiming Shudian.
- Qian, M. (1952). *Zhongguo lidai zhengzhi deshi* [The Political Gains and Losses of Chinese Dynasties]. Self-published, Hong Kong.

Hsiao, K. (1960). *Rural China: Imperial Control in the Nineteenth Century*. University of Washington Press.

Mao, H. (1995). *Tianchao de bengkui: Yapien zhanzheng zai yanjiu* [The Collapse of the Celestial Empire: A Reexamination of the Opium War]. Sanlian Shudian.

Kuhn, P. A. (1990). *Soulstealers: The Chinese Sorcery Scare of 1768*. Harvard University Press.

Western History

Brunt, P. A. (1971). *Social Conflicts in the Roman Republic*. W. W. Norton & Company.

Brunt, P. A. (1988). *The Fall of the Roman Republic and Related Essays*. Clarendon Press.

Syme, R. (1939). *The Roman Revolution*. Oxford University Press.

Gruen, E. S. (1974). *The Last Generation of the Roman Republic*. University of California Press.

Millar, F. (1998). *The Crowd in Rome in the Late Republic*. University of Michigan Press.

Plutarch. (c. 100 CE). *Bioi parallēloi* [Parallel Lives]. (B. Perrin, Trans., 1914–1926, Loeb Classical Library. Harvard University Press). (Especially "Tiberius Gracchus" and "Gaius Gracchus").

Appian. (c. 150 CE). *Rhōmaiká* [Roman History]. (H. White, Trans., 1899, Loeb Classical Library. Macmillan). (Especially the "Civil Wars" books).

Brown, A. (2009). *The Rise and Fall of Communism*. Ecco.

Kotkin, S. (2001). *Armageddon Averted: The Soviet Collapse, 1970–2000*. Oxford University Press.

Gorbachev, M. (1996). *Memoirs*. Doubleday.

VII. Information Ecology and Contemporary Cognitive Governance

Zuboff, S. (2019). *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. PublicAffairs.

Pariser, E. (2011). *The Filter Bubble: What the Internet Is Hiding from You*. Penguin Press.

Sunstein, C. R. (2001). [Republic.com](http://www.republic.com). Princeton University Press.

Benkler, Y. (2006). *The Wealth of Networks: How Social Production Transforms Markets and Freedom*. Yale University Press.

Castells, M. (1996–1998). *The Information Age: Economy, Society, and Culture* (Vols. 1–3). Blackwell.

Floridi, L. (2014). *The Fourth Revolution: How the Infosphere is Reshaping Human Reality*. Oxford University Press.

Han, B.-C. (2017). *In the Swarm: Digital Prospects*. MIT Press.

VIII. Philosophy of Action and Theories of Transformation

Freire, P. (1970). *Pedagogia do Oprimido* [Pedagogy of the Oppressed]. Herder and Herder.

Arendt, H. (1958). *The Human Condition*. University of Chicago Press.

Arendt, H. (1963). *On Revolution*. Viking Press.

Gramsci, A. (1929–1935). *Quaderni del carcere* [Prison Notebooks]. (J. A. Buttigieg, Ed. & Trans., 1992–2007, Columbia University Press).

Berger, P. L., & Luckmann, T. (1966). *The Social Construction of Reality: A Treatise in the Sociology of Knowledge*. Doubleday.

Giddens, A. (1984). *The Constitution of Society: Outline of the Theory of Structuration*. Polity Press.

Archer, M. S. (1995). *Realist Social Theory: The Morphogenetic Approach*. Cambridge University Press.

Illich, I. (1971). *Deschooling Society*. Harper & Row.

Castoriadis, C. (1975). *L'institution imaginaire de la société* [The Imaginary Institution of Society].

Seuil.

IX. Supplementary Theoretical Resources

Bateson, G. (1972). *Steps to an Ecology of Mind*. Chandler Publishing.

Bateson, G. (1979). *Mind and Nature: A Necessary Unity*. E. P. Dutton.

Maturana, H. R., & Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel.

Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.

Deleuze, G., & Guattari, F. (1980). *Mille plateaux [A Thousand Plateaus]*. Minuit.

Foucault, M. (1975). *Surveiller et punir: Naissance de la prison [Discipline and Punish: The Birth of the Prison]*. Gallimard.

Foucault, M. (1966). *Les mots et les choses [The Order of Things]*. Gallimard.

Bibliographic Note

This bibliography serves the programmatic general treatise of the “Civilization Phase Transition Dynamics” research program. The works cited are organized into nine categories:

1. **Author’s Core Theoretical Works:** The direct theoretical foundation of this paper, containing the systematic exposition of all axioms, theorems, and diagnostic criteria.
2. **Civilizational Morphology and Philosophy of History (Old Paradigm Representatives) :** The primary objects of this paper’s critique, representing the tradition of cyclical historiography from classical antiquity to the modern era.
3. **Complex Systems and Phase Transition Theory:** Provides the natural-scientific foundations for the concept of “phase transition,” especially theories of self-organized criticality, percolation models, and minority influence.
4. **Cognitive Science and Metacognition Theory:** Provides the cognitive-scientific foundations for Axiom 1 (Logical Self-Reference), Axiom 4 (Relativity of Frameworks), Axiom 5 (Symbol as Noospheric Gene), and the Cognitive Escape Theorem.
5. **Systems Theory and Social Systems Pathology:** Provides the systems-theoretic and social-scientific dialogue partners for concepts such as systemic parasites, functional alienation, and civilizational health.
6. **Chinese and Western Historical Case Studies:** Supports the comparative case analyses in Parts Four and Five, covering the core cases of the Roman Republic, Northern Song, Mid-Ming, Late Qing, and the Soviet Union.
7. **Information Ecology and Contemporary Cognitive Governance:** Provides the empirical background and critical resources for the dynamical diagnosis of contemporary civilization in Part Six.
8. **Philosophy of Action and Theories of Transformation:** Provides the philosophical and practical-theoretical dialogue partners for the program of action for awakened individuals.
9. **Supplementary Theoretical Resources:** Provides extended reading for the deeper philosophical dimensions of the theory, including reflexivity and the ontology of the noospheric ecology.

Within the main text, only the core works in the first two categories are directly cited. The remaining works serve as theoretical background and dialogue partners, offered to the reader for further exploration of the academic lineages intersecting with this paper’s arguments.

